Class 6
Physics

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1. Units and Measurements

Level – I

1. **Answer: A.** Anything that can be measured is known as a physical quantity.
2. **Answer: B.** The set of length, time, and velocity cannot enter the fundamental list because velocity is derived mathematically from length and time, meaning they aren't totally independent of each other.
3. **Answer: B.** Mass is the only measurable physical quantity listed. Feelings like sadness or affection simply can't be measured with physical units.
4. **Answer: A.** Physical quantities that are completely independent of others are called fundamental physical quantities.
5. **Answer: B.** A quantity that requires other quantities to be derived is called a derived quantity.
6. **Answer: A.** If you multiply 1 centimetre by 10, you get 10 millimetres.
7. **Answer: D.** Since there are 10 millimetres in a centimetre, 1 millimetre is exactly 1/10 of a centimetre.
8. **Answer: B.** One quintal is equal to 100 kilograms, which can be written in exponential form as 10^2 kg.
9. **Answer: B.** There are exactly 1000 metres in 1 kilometre.
10. **Answer: D.** To convert metres to centimetres, you multiply by 100. So, 0.01 metres multiplied by 100 gives you 1 centimetre.
11. **Answer: A.** The prefix nano means one-billionth, making 1 nanometre equal to 10^{-9} metres.
12. **Answer: B.** A century represents a block of 100 years.
13. **Answer: D.** The prefix pico means one-trillionth, so 1 picometre equals 10^{-10} centimetres.
14. **Answer: B.** A standard quantity used for comparison
15. **Answer: A.** The number part that tells you how many times a standard unit fits into a measurement is called the numerical value.
16. **Answer: B.** Kelvin is the fundamental SI unit used to measure temperature.
17. **Answer: B.** Energy is mathematically derived from mass, length, and time, meaning it is not a fundamental quantity.
18: **Answer: C.** Since nano stands for 10^{-9}, 400 nm is written in scientific notation as 400×10^{-9} metres.
19: **Answer: C.** To make sense of any physical measurement, a unit is required for comparison.
20. **Answer: A.** The prefix micro stands for one-millionth, meaning one microgram equals 10^{-6} grams.
21. **Answer: A.** One standard Joule of energy perfectly equals ten million (10^7) ergs.
22. **Answer: B.** There are 60 seconds in one minute. Multiplying 40 minutes by 60 seconds per minute gives a total of 2400 seconds.

Level – II

Single Option Correct Type

1. **Answer: B.** Out of the given options, the centimetre is the most appropriate unit for measuring smaller lengths.
2. **Answer: D.** The standard symbol used for the amount of a substance is Mol.
3. **Answer: C.** The S.I. unit for the strength of an electric current is the Ampere.

4. **Answer: D.** The physical quantity stays the same no matter the unit, so the first numerical value multiplied by its unit will equal the second numerical value multiplied by its unit. This makes the equation $n_1U_1 = n_2U_2$.
5. **Answer: D.** The standard S.I. unit of luminous intensity is the Candela.
6. **Answer: D.** This is a duplicate of 2. The symbol is Mol.
7. **Answer: A.** To convert a speed from metres per second to kilometres per hour, you multiply by the fraction 18/5. This gives the equation $y = x \times 18/5$. Rearranging this to find the ratio x/y gives you exactly 5/18.

One or More Than One Option Correct Type

8. **Answer: A and B**. One metric ton is exactly equal to 1000 kilograms. Since 1 quintal is 100 kilograms, dividing 1000 by 100 tells us that 1 metric ton is also equal to 10 quintals.
9. **Answer: A, B, C, D:** Since one microgram is equal to 10^{-6} grams, 5 micrograms is exactly 5×10^{-6} **g** (Option B). To find the weight in kilograms, you multiply the gram value by 10^{-3}, which gives 5×10^{-9} **kg** (Option A). Converting those kilograms to metric tonnes requires multiplying by another 10^{-3}, resulting in 5×10^{-12} **tonnes** (Option C). Finally, because there are 1,000 micrograms in a single milligram, dividing the original 5 micrograms by 1,000 yields 5×10^{-3} **mg** (Option D).

Comprehension Type

10. **Answer: B, C.** Mass and Length are completely independent fundamental physical quantities.
11. **Answer: C and D**. Both acceleration and density depend on other fundamental quantities like mass, length, and time, making them derived quantities.
12. **Answer: A.** Deca means ten times the unit, making a decametre a multiple of a metre. The other options are submultiples used for smaller measurements.
13. **Answer: B.** A millimetre represents a fraction of a metre, making it a true sub-multiple.
14. **Answer: B.** A milligram is a sub-multiple used to measure tiny amounts of mass smaller than a gram.
15. **Answer: A.** The size of the unit and the numerical value act like a seesaw. If the numerical value is smaller, the unit measuring it must be relatively large.
16. **Answer: A.** Based on the same concept as the previous , the relation between a numerical value and its unit is inversely proportional.

Assertion and Reason Type

17. **Answer: A**. Assertion (A) is considered true because you cannot properly express a physical quantity without a unit. Reason (R) perfectly explains why: if you only give a number like "10," no one knows if you mean 10 meters, 10 kilograms, or 10 seconds. The unit is absolutely necessary to give the numerical value its meaning..
18. **Answer: A.** Both the assertion and the reason are completely true. A standard unit must be invariable, and the reason gives the perfect explanation why—it shouldn't change under any circumstances.

Matrix Matching Type

19. **a→T, b→R, c→Q, d→P:** Micro matches with T because it means 10^{-6}. Centi matches with R because it means 10^{-2}. Giga matches with Q because it means 10^{9}. Mega matches with P because it means 10^{6}.
20. **a→Q, b→R, c→Q, d→R:** Thermodynamic temperature matches with Q because it is a fundamental quantity. Density matches with R because it is a derived quantity.

Mass matches with Q because it is a fundamental quantity. Volume matches with R because it is derived from length.

Integer Type

21. **The answer is 3**. To convert centimetres to metres, you divide by 100. So, 300 centimetres divided by 100 equals 3 metres.

22. **The answer is 1000.** One tonne is defined as exactly 1000 kilograms.

23. **The answer is 1000.** A metric ton is the same measurement as a tonne, meaning it also equals 1000 kilograms.

Measurements

Level – I

1. **Answer: B** The volume of a sphere depends on its radius because the mathematical formula requires the radius to calculate it.

2. **Answer: C** Just multiply the room's dimensions to get a volume of 50 cubic meters, then multiply by the air density of 1.3 to get a mass of 65 kg.

3. **Answer: B** Multiply the length, breadth, and height together to get a total volume of 3.75 cubic meters.

4. **Answer: D** Find the area of both paths and subtract the overlapping square in the middle to get 45 square meters.

5. **Answer: A** The circle's area is 616, and the biggest square you can fit inside has an area of 392, leaving a remaining area of 224.

6. **Answer: B** Square the radius of 2.1 and multiply it by pi to get an area of 13.86 square meters.

7. **Answer: B** Change the height to 2.1 cm, then plug it into the cone volume formula to get 35.2 cubic centimeters.

8. **Answer: B** Find the cylinder's volume which is 1331, and since the new cube holds the exact same amount, its side must be 11 cm.

9. **Answer: A** Since one hectare is exactly 10000 square meters, fifty hectares becomes 500000 square meters.

10. **Answer: A** Use the triangle area formula of half times the base times the height to get 20 square meters.

11. **Answer: A** Square the radius of 200 and multiply by pi to get a total area of 40000 pi.

12. **Answer: C** Multiply the side length of 10 cm by 4 to easily find the perimeter of 40 cm.

13. **Answer: B** Use the circle area formula with a radius of 7 to get 154 square centimeters.

14. **Answer: A** Divide the total area of 120 by the breadth of 6 to find the missing length of 20 cm.

15. **Answer: B** Add the length and breadth to get 15, then double it to find the total perimeter of 30 cm.

16. **Answer: C** A measuring jar is the perfect tool for directly reading how much liquid volume you have.

17. **Answer: B** The standard CGS unit used to express volume is cubic centimeters.

18. **Answer: B** One single milliliter is exactly equal to one cubic centimeter.

19. **Answer: B** The physical space that any liquid takes up inside a container is simply called its volume.

20. **Answer: B** Use the standard total surface area formula for a cuboid to find the answer of 310 square meters.

21. **Answer: B** A pipette is specifically designed for measuring out a fixed, precise amount of liquid.
22. **Answer: B** There are exactly 10 millimeters packed into one single centimeter.
23. **Answer: D** kilometers are generally used for measuring distance between two cities
24. **Answer: C** A square with sides of exactly 1 meter will logically have a total area of 1 square meter.
25. **Answer: C** A hectare is a standard unit for land measurement that equals 10000 square meters.
26. **Answer: C** Square kilometers are generally the best choice for measuring huge pieces of land.

Level – II

1. **Answer: B** Just multiply the length of 4 meters by the breadth of 3 meters to get an area of 12 square meters.
2. **Answer: B** Change the centimeters to nanometers by multiplying by ten million, giving you the correct converted number.
3. **Answer: B** Based on the dimensions provided for the rectangle, multiplying length by width gives 16 square meters plus area of triangle 2 square meters total 18 square meters.
4. **Answer: B** You can easily find a cuboid's volume by multiplying its length, breadth, and height together.
5. **Answer: B or C** One decagram equals 10 grams, which mathematically converts to ten to the power of negative two kilograms.
6. **Answer: A** Since one meter contains 100 cm, squaring that gives you a total of 10000 square centimeters.
7. **Answer: B** The number of unit squares that fit perfectly inside a flat space is defined as its area.
8. **Answer: B** Perimeter always refers to the total length around the outside edge or boundary of a shape.
9. **Answer: C** The officially recognized SI unit used for measuring volume is the cubic meter.
10. **Answer: C** There are exactly one thousand milliliters contained within one liter.
11. **Answer: C** Dropping a strangely shaped object into water and checking how much the water level rises is the best way to measure it.
12. **Answer: C** A container with a space of exactly one cubic meter holds 1000 liters.

One or More Than One Option Correct Type

13. **Answer: A**, **B, D** Volume describes the space occupied, and its correct MKS unit is indeed cubic meters. CGS units of area is square centimeters.
14. **Answer: A, B, C** The area formulas for rectangles and squares are correct, and regular rulers measure down to millimeters.
15. **Answer: A, B, C** A liter is 1000 ml, one cubic meter holds 1000 liters, and one cubic centimeter is exactly a millionth of a cubic meter.
16. **Answer: A, B, C, D** All of these options represent totally valid units for measuring liquid volume.

Comprehension Type

17. **Answer: B** Just divide 50 centimeters by 100 to change it straight into 0.5 meters.
18. **Answer: B** Take away the old water level from the new one to find out the object's volume is 175 mL.

19. **Answer: A** Multiply the 2 meter length by the 0.5 meter breadth to easily find the area is 1 square meter.

20. **Answer: C** Multiply the 10 cm side by itself three times to get a volume of 1000 cubic centimeters.

21. **Answer: B** Since 1000 cubic centimeters translates exactly to one liter, the volume is simply 1 L.

22. **Answer: C** Because the cube's volume is exactly 1 liter and you are comparing it to 1 liter of water, both amounts are completely equal.

Assertion and Reason Type

23. **Answer: A** Both the statements are true, and the reasoning about squaring the conversion cleanly explains why a square meter is 10000 square cm.

24. **Answer: A** The given statements are correct, and the volume formula logic completely explains the conversion into cubic centimeters.

Matrix Matching Type

25. **Answer: A-Q, b-S, c-R, d-P** This directly matches each specific geometric shape to its correct standard formula and calculated values.

26. **Answer: A-S, b-R, c-Q, d-P** This matches every three-dimensional shape provided straight to its widely known volume formula.

Integer Type

27. **Answer: 250** Subtract 400 from 650 to clearly see that the water went up by 250 mL, which is your object's volume.

28. **Answer: 1000** Multiply the side length of 10 by itself three times to calculate the total volume in cubic centimeters.

29. **Answer: 1** Convert the 50 cm breadth to 0.5 m, then multiply it by the length of 2 m to get a clean area of 1.

Previous Year Questions

1. **Answer: D** Just subtract the starting measurement of 5 from the final reading of 14.2 to get the accurate length.

2. **Answer: D** Figure out the full length of the wrapped thread on the ruler and divide by 6 turns to see each turn needs 2.03 cm.

3. **Answer: A** Since Trisha has to travel a shorter distance of 500 meters compared to Ansh's 600 meters, she naturally gets there first.

4. **Answer: C** Looking at the pencil straight on gives an actual length of 30 mm, while viewing it from an angle creates parallax making it seem longer.

5. **Answer: D** A standard meter splits into 100 equal centimeters, and each centimeter breaks down further into 10 little millimeters.

6. **Answer: D** When checking the math on these conversions, only the first and fourth options actually make sense.

7. **Answer: C** Change the wire length of 18 cm into millimeters, then divide it by the tiny length of a single turn to find there are 4500 total turns.

2. Kinematics

Level – I

1. **Answer:** B) Relative
Rest and motion depend on the observer's frame of reference. An object may be at rest relative to one observer while in motion relative to another, making them relative terms.

2. **Answer:** A) Periodic
The moon completes its orbit around the earth in a fixed interval of time (approximately 27.3 to 30 days). Motion that repeats at regular time intervals is called periodic motion.

3. **Answer:** B) Rectilinear
When a body moves strictly in a straight line, its motion is classified as rectilinear motion.

4. **Answer:** A) Periodic
A simple pendulum swings back and forth in equal intervals of time. Because the motion repeats in a fixed time interval, it is periodic motion.

5. **Answer:** A) Rotatory
A potter's wheel spins around a fixed central axis without changing its overall position. This is the definition of rotatory (or rotational) motion.

6. **Answer:** B) Curvilinear
When an athlete throws a javelin, it travels through the air in a curved path under the influence of gravity (projectile motion), which is a type of curvilinear motion.

7. **Answer:** C) Rotatory
The blades of a windmill turn around a fixed central axis, executing rotatory motion.

8. **Answer:** A) The needle of the sewing machine is in an oscillatory motion
In a sewing machine, the needle moves up and down continuously, which is oscillatory (vibratory) motion. The wheel, however, undergoes rotatory motion.

9. **Answer:** B) Rotatory
If all particles of an object move in concentric circular paths around a single fixed axis, the object as a whole is in rotatory motion.

10. **Answer:** A) Rotatory
While moving on a straight road, the cycle itself is in translatory motion, but the pedal moves in a circular path around its axle, possessing rotatory (circular) motion.

11. **Answer:** D) Oscillatory
The pendulum of a clock moves to and fro about a mean position, which is the definition of oscillatory motion.

12. **Answer:** A) Oscillatory
The piston of an engine moves back and forth (reciprocates) linearly inside a cylinder. This to-and-fro motion is a form of oscillatory motion.

13. **Answer:** B) Oscillatory
When the strings of a violin are plucked or bowed, they vibrate back and forth rapidly about their resting position, which is oscillatory (vibratory) motion.

14. **Answer:** A) A car taking a turn
A car taking a turn moves along a curved path, which is curvilinear motion. The other options describe rectilinear motion.

15. **Answer:** A) Scalar
Distance has only magnitude and no specific direction, making it a scalar quantity.

16. **Answer:** B) Metre
The standard International System (S.I.) unit for measuring distance or length is the metre.

17. **Answer:** A) Initial and final position of the body
Displacement is defined as the shortest straight-line distance between the initial position and the final position of a moving body.

18. **Answer:** C) Zero
The person starts at point P, travels along the path PQRP, and ends exactly back at point P. Since the initial and final positions are identical, the displacement is zero.

19. **Answer: Invalid Question**
If we assume the object moves along two perpendicular sides of a right-angled triangle (e.g., 1 m and 2 m), the shortest path (displacement) is calculated using the Pythagorean theorem:
$Displacement = \sqrt{1^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5}$ m.

20. **Answer:** B) Odometer
The distance traveled by a vehicle is measured by an instrument on the dashboard called an odometer. A speedometer measures instantaneous speed.

21. **Answer:** B) πr
The circumference of a full circle is given by $2 \times \pi \times r$. For half a circle, the distance covered is half of the circumference:
$Distance = \frac{2 \times \pi \times r}{2} = \pi r$.

22. **Answer:** D) Equal to 1 or less than 1
Displacement is the shortest distance between two points. It can never be greater than the actual distance traveled. Therefore, the ratio of displacement to distance is always ≤ 1.

23. **Answer:** B) 13 m
The horse runs 12 m North and 5 m East, forming a right-angled triangle. The displacement is the hypotenuse.
$Displacement = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$ m.

24. **Answer:** B) 0 km
In a round trip, the starting point and ending point are the same (home). Therefore, the overall change in position, or displacement, is 0 km.

25. **Answer:** B) $2\pi r$, 0
For one complete revolution in a circular path of radius r, the distance is the full circumference ($2\pi r$). The displacement is 0 because the particle returns to its starting point.

26. **Answer:** B) $10\pi R$
The distance for one revolution is $2\pi R$. For 5 revolutions, the total distance is:
$Distance = 5 \times 2\pi R = 10\pi R$.

27. **Answer:** D) 10 m
The particle moves 8 m East and 6 m North. These paths are perpendicular.
$Displacement = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$ m.

28. **Answer:** D) It must be zero
If the distance covered is zero, it means the particle has not moved at all from its initial position. Thus, its displacement must also be zero.

29. **Answer:** B) 2h, 0
The body travels a distance 'h' upwards and a distance 'h' downwards.
Total distance = $h + h = 2h$.
Since it returns to the ground (its starting point), the displacement is 0.

30. **Answer:** C) 15π m, 0
The diameter is 5 m. Distance for one revolution is $\pi \times d = 5\pi$ m.
For 3 revolutions: $Distance = 3 \times 5\pi = 15\pi$ m.
Since it completes full revolutions, it returns to the start,
making displacement 0.

31. **Answer:** D) Simple pendulum
A simple pendulum exhibits periodic motion (each swing takes the same amount of time), which is the principle historically used in clocks to measure time. Hand span, stride, and foot are units of length.

32. **Answer:** B) 2400
There are 60 seconds in one minute.
40 minutes = 40 × 60 seconds = 2400 seconds.

33. **Answer:** B) February
In a standard calendar year, February has 28 days. In a leap year, an extra day is added, giving February 29 days.

34. **Answer:** C) 1.5
There are 60 seconds in a minute. To convert seconds to minutes, divide by 60.
90 seconds = $\frac{90}{60}$ minutes = 1.5 minutes.

35. **Answer:** B) 3000
There are 1000 milliseconds (ms) in one second.
3 seconds = 3 × 1000 ms = 3000 ms.

Level – II

SINGLE OPTION CORRECT TYPE

1. **Answer:** B) Spinning top around its own axis
A swinging cradle is oscillatory, house flies move randomly, and a car in a straight line is in translatory motion. A spinning top moves around a fixed axis, which is rotatory (where particles execute circular motion).

2. **Answer:** A) At rest
By definition, if an object's position does not change relative to its surroundings over time, it is said to be at rest.

3. **Answer:** C) Time
Motion is defined as the change in position of an object with respect to its surroundings and with respect to time.

4. **Answer:** A) Curvilinear
A kicked football travels in a parabolic arc due to gravity, which is a curve.
Therefore, it is an example of curvilinear motion.

5. **Answer:** B) Circular motion
Circular motion is a specific type of curvilinear/rotatory motion where a particle moves along a path such that its distance from a central fixed point (the radius) remains strictly constant.

6. **Answer:** B) 14 m
Distance is the total path length traveled regardless of direction.
$Distance$ = 8 m (East) + 6 m (North) = 14 m.

7. **Answer:** A) 13 m, 5 m
Total distance = 3 m + 4 m + 6 m = 13 m.
To find displacement, calculate the net distances on coordinate axes:
Net North-South = 3 m North - 6 m South = 3 m South.
Net East-West = 4 m East.
$Displacement = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ m.

8. **Answer:** A) Displacement zero
In one hour, the minute hand completes exactly one full revolution. It returns to its starting point, so displacement is zero. The tip covers a path equal to the circumference, so the distance covered is $2\pi R$ not $2\pi r$.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. **Answer:** A) Rotatory motion and B) Translatory motion
A rolling ball physically advances forward (translatory motion) while simultaneously spinning around its center axis (rotatory motion).

10. **Answer:** A) Positive and C) Zero
Magnitude refers to the absolute value or size of a physical quantity, which can never be strictly negative. It is either zero (if there is no net change in position) or positive.

11. **Answer:** A, B, and D
The hands of a watch, the rotation of the earth on its axis, and the revolution of the earth around the sun all repeat at fixed, regular intervals of time, making them periodic.

12. **Answer:** A, B, C, and D
A spinning top, windmill, ceiling fan, and car wheels all turn about a fixed axis, which means they all execute rotatory motion.

13. **Answer:** B) Translatory motion
When a car moves on a straight, level road, the entire body of the car shifts uniformly in one direction, which defines translatory (specifically rectilinear) motion.

14. **Answer:** A) 100 m
One trip implies the crow leaves the pot, goes 10 m to get a pebble, and returns 10 m to the pot.
Distance per trip = 10 m + 10 m = 20 m.
Total distance for 5 trips = 5 × 20 m = 100 m.

ASSERTION AND REASONING

15. **Answer:** A) Both A and R are true and R is the correct explanation of A
A chair in a room is at rest relative to the room because its position is not changing with respect to its immediate surroundings. The reason correctly defines the assertion.

16. **Answer:** A) Both A and R are true and R is the correct explanation of A
Vibratory motion is indeed a subcategory of oscillatory motion, characterized specifically by a very rapid to-and-fro movement.

COMPREHENSION TYPE

17. **Answer:** B) 800 m
The length of the track is 200 m for one round.
Total distance for 4 rounds = 4 × 200 m = 800 m.

18. **Answer:** D) 0 m
Because the athletes complete exact rounds and the line of finish is the same as the line of start, their final position is identical to their initial position, making displacement 0.

19. **Answer:** D) Non uniform
In a long-distance race, an athlete speeds up, slows down, and gets tired, meaning they do not cover equal distances in equal intervals of time. Hence, the motion is non-uniform.

20. **Answer:** B) Not equal
The total distance is 800 m, whereas the displacement is 0 m. Therefore, distance and displacement are not equal.

21. **Answer:** B) Relative
As stated in the passage, an object can be at rest for one observer and in motion for another, proving that rest and motion are relative concepts.

22. **Answer:** C) Each other
Passengers sitting inside the same moving bus are not changing positions relative to one another, so they are at rest with respect to each other.

23. **Answer:** C) The observer or reference frame
The condition of being at rest or in motion completely depends on the point of view of the person observing, which is called the frame of reference.

MATRIX MATCHING TYPE

24. **Answer: a-R, b-Q, c-S, d-P**
A swinging cradle moves to and fro (Oscillatory). A spinning top rotates on its axis (Rotatory). Fishes move unpredictably in water (Random). A car on a straight road moves linearly (Rectilinear).

25. **Answer: a-Q, b-T, c-S, d-P**
Distance is the total path traveled (Q). Displacement is the shortest straight line joining initial and final positions (T). Distance in one circular rotation matches the formula for circumference (S - representing one rotation). Displacement in one full rotation is zero (P).

26. **Answer: a-S, b-P or T, c-Q, d-R**
1 hour equals 3600 seconds (S). 1 day consists of 24 hours (P) = 86400 seconds (T). 1 decade is a period of 10 years (Q). A century is a period of 100 years (R).

INTEGER TYPE

27. **Answer:** 86154
A mean solar day is 24 hours, which equals $24 \times 60 \times 60 = 86400$ seconds.
A sidereal day is 4.1 minutes shorter.
4.1 minutes $= 4.1 \times 60 = 246$ seconds.
Duration of sidereal day $= 86400 - 246 = 86154$ seconds.

28. **Answer:** 259200000
1 solar day = 24 hours = 86400 seconds.
3 solar days $= 3 \times 86400$ seconds $= 259200$ seconds.
To convert to milliseconds, multiply by 1000:
$259200 \times 1000 = 259200000$ ms.

29. **Answer:** 40
A square has 4 sides. If the particle returns to the starting point, it traveled all 4 sides.
Distance $= 4 \times 10$ m $= 40$ m.

Since it returns to the starting point, displacement = 0 m.
Sum = 40 + 0 = 40.

Speed, Velocity, and Acceleration

LEVEL – I

1. **Answer:** C) 40 m
Solution: The initial speed is given as 72 km/h. Using the conversion factor, the speed is $72 \times \frac{5}{18} = 20$ m/s. The time given is 2 sec. Using the formula Distance = Speed × Time, the calculation is $20 \times 2 = 40$ m.

2. **Answer:** A) Distance / Time
Solution: The formula for speed is defined as the total distance traveled divided by the total time taken. Therefore, the correct relation is
Speed = $\frac{\text{Distance}}{\text{Time}}$.

3. **Answer:** B) $6\ ms^{-1}$
Solution: Let the total distance be $2d$. The time taken for the first half is $t_1 = \frac{d}{2}$ and for the second half is $t_2 = \frac{d}{v}$. The formula for average speed is Average Speed = $\frac{\text{Total Distance}}{\text{Total Time}}$. Substituting the given average speed of 3 ms^{-1}, we get $3 = \frac{2d}{\frac{d}{2}+\frac{d}{v}}$.
Simplifying this equation gives $\frac{1}{2}+\frac{1}{v}=\frac{2}{3}$, which ultimately results in $v = 6$ ms^{-1}.

4. **Answer:** C) $100\ m/s$
Solution: The train covers a displacement of 200 m towards the west in 2 s. Using the formula Velocity = $\frac{\text{Displacement}}{\text{Time}}$, the calculation is $\frac{200}{2} = 100$ m/s towards the west.

5. **Answer:** A) $1.25\ m/s$
Solution: The distance to the pebbles is 10 m. A single trip (going and returning) covers 20 m. For 5 trips, the total distance is $5 \times 20 = 100$ m. The total time taken is 80 seconds. Using the formula Average Speed = $\frac{\text{Total Distance}}{\text{Total Time}}$, the calculation is $\frac{100}{80} = 1.25$ m/s.

6. **Answer:** B) Odometer
Solution: An odometer is the standard instrument used in vehicles that measures and records the total distance moved by the vehicle

7. **Answer:** C) $5\ ms^{-1}$
Solution: The body travels a distance of 10 m in 2 s. Using the formula Speed = $\frac{\text{Distance}}{\text{Time}}$, the calculation is $\frac{10}{2} = 5$ ms^{-1}.

8. **Answer:** B) 20 minutes
Solution: The total distance is 20 km and the constant speed is 60 km/h. Using the formula Time = $\frac{\text{Distance}}{\text{Speed}}$, the calculation is $\frac{20}{60}$ hours, which equals $\frac{1}{3}$ of an hour. Converting to minutes, $\frac{1}{3} \times 60 = 20$ minutes.

9. **Answer:** A) Total distance / Total time
Solution: The mathematical definition for average speed is the ratio of the total path length traveled to the total time interval taken.

10. **Answer:** A) $10\ km\ h^{-1}$
Solution: The body moves with a uniform speed of 10 km/h for 2 h. Because the speed is constant (uniform) throughout the journey, the average speed remains equal to the uniform speed, which is 10 km/h.

11. **Answer:** B) Vector
Solution: Velocity indicates both the speed of an object and its direction of motion, meaning it possesses both magnitude and direction, classifying it as a vector quantity.

12. **Answer:** C) 50 km/h (Exact answer is 48 km/h nearly 50 km/h)
Solution: The car travels 100 km at 60 km/h and returns 100 km at 40 km/h. The total distance is 200 km. The total time is $\frac{100}{60}+\frac{100}{40}=\frac{5}{3}+\frac{5}{2}=\frac{25}{6}$ hours. Using the formula
Average Speed $=\frac{\text{Total Distance}}{\text{Total Time}}$,
the exact calculation is $\frac{200}{\frac{25}{6}}=48$ km/h.

13. **Answer:** A) Its instantaneous speed
Solution: The speedometer of a car is designed to measure and indicate the speed of the vehicle at that exact moment in time, which is known as instantaneous speed.

14. **Answer:** A) $-2m/s^2$
Solution: The initial velocity u is 10 m/s and the final velocity v is 0 m/s since the car comes to rest in $t=5$ s. Using the formula $a=\frac{v-u}{t}$, the calculation is $\frac{0-10}{5}=-2$ m/s^2.

15. **Answer:** B) cm/s^2
Solution: In the Centimeter-Gram-Second (C.G.S) system, length is measured in centimeters and time in seconds. Therefore, the unit for acceleration, which is distance per unit time squared, is cm/s^2.

16. **Answer:** A) Zero
Solution: Uniform velocity means the velocity is constant and not changing over time. Since acceleration is defined as the rate of change of velocity, a constant velocity results in an acceleration of zero.

17. **Answer:** A) Retardation
Solution: When the velocity of an object decreases over time, it undergoes negative acceleration. This specific decrease in velocity per unit time is commonly called retardation or deceleration.

18. **Answer:** C) $1m/s^2$
Solution: The scooter acquires a final velocity of 36 km/h. Converting to standard units: $36\times\frac{5}{18}=10$ m/s. The initial velocity is 0 and the time is 10 s. Using the formula $a=\frac{v-u}{t}$, the calculation is $\frac{10-0}{10}=1$ m/s^2.

19. **Answer:** D) Either the speed or direction or both change
Solution: Velocity is a vector quantity, meaning it is defined by both its magnitude (speed) and its direction. Thus, velocity is considered variable if there is a change in speed, a change in direction, or a change in both simultaneously.

20. **Answer:** B) $3ms^{-1}, 0$
Solution: Usha covers a total distance of 180 m in 1 minute (60 s). Using the formula Average Speed $=\frac{\text{Total Distance}}{\text{Total Time}}$, the calculation is $\frac{180}{60}=3$ ms^{-1}. Since she returns to her starting point, her net displacement is 0 m. Thus, Average Velocity $=\frac{\text{Displacement}}{\text{Time}}=\frac{0}{60}=0$ ms^{-1}.

21. **Answer:** C) Velocity
Solution: Motion in a specified direction refers to the change in an object's position (displacement). The rate of change of this directed motion is defined as velocity.

22. **Answer:** A) Uniform acceleration
Solution: When a body experiences equal changes in its velocity during equal intervals of time, regardless of how small those intervals are, its rate of change of velocity is constant, which is the definition of uniform acceleration.

23. **Answer:** B) 18s
Solution: The car starts from rest, meaning initial velocity $u = 0$, and accelerates at $a = 3\ \text{m/s}^2$ to reach a final velocity $v = 54$ m/s. Using the formula $t = \frac{v-u}{a}$, the calculation is $\frac{54-0}{3} = 18$ s.

24. **Answer:** C) $0.216\ km/h$
Solution: The speed is given as 0.06 m/s. To convert from m/s to km/h, multiply by $\frac{18}{5}$ (or 3.6). The calculation is $0.06 \times 3.6 = 0.216$ km/h.

25. **Answer:** B) Acceleration
Solution: The term used to describe the continuous increase in the velocity of a moving body over time is acceleration, which represents the rate of change of velocity.

26. **Answer:** C) In accelerated motion
Solution: Although the boy is moving with a constant speed of 10 ms^{-1} on the merry-go-round, his direction of motion is continuously changing due to the circular path. Since velocity includes direction, his overall velocity is continuously changing, meaning he is in accelerated motion.

27. **Answer:** C) $v < u$
Solution: Negative acceleration (retardation) occurs when an object is slowing down. This means the final velocity v is less than the initial velocity u, resulting in the condition $v < u$.

28. **Answer:** B) $2ms^{-2}, +1ms^{-2}$
Solution: First, convert the velocities: 36 km/h = 10 m/s and 72 km/h = 20 m/s . The first acceleration is $a_1 = \frac{20-10}{5} = 2\ \text{ms}^{-2}$. When stopping from 20 m/s in 20 seconds, the final velocity is 0. The new acceleration is $a_2 = \frac{0-20}{20} = -1\ \text{ms}^{-2}$. Retardation is the negative of acceleration = $+1\ \text{ms}^{-2}$.

LEVEL – II

1. **Answer:** A) $\pi m/s$
Solution: The particle moves in a semicircle of radius $r = 2$ m. The total distance traveled is half the circumference, calculated as $\pi \times r = 2\pi$ m. The time taken is 2 seconds. Using the formula Speed = $\frac{\text{Distance}}{\text{Time}}$, the calculation is $\frac{2\pi}{2} = \pi$ m/s.

2. **Answer:** D) $\frac{2\pi r}{t}$
Solution: For a complete rotation along a circular path of radius r, the total distance covered is the full circumference of the circle, which is $2\pi r$. If the time taken is t, then using the formula Speed = $\frac{\text{Distance}}{\text{Time}}$, the average speed is $\frac{2\pi r}{t}$.

3. **Answer:** A) π
Solution: The particle travels along a semicircle of radius $r = 1$ m. The distance

covered is $\pi \times r = \pi \times 1 = \pi$ m. The time taken is 1 sec. Using the formula Average Speed $= \frac{\text{Distance}}{\text{Time}}$, the calculation is $\frac{\pi}{1} = \pi$ m/s.

4. **Answer:** C) 2.5

Solution: The initial velocity is 108 km/h, which converts to $108 \times \frac{5}{18} = 30$ m/s. The final velocity is 72 km/h, converting to $72 \times \frac{5}{18} = 20$ m/s. The time taken is 4 s. Using the formula for acceleration $a = \frac{v-u}{t}$, the calculation is $\frac{20-30}{4} = -2.5$ m/s^2. The deceleration is represented by the absolute magnitude, which is 2.5 m/s^2.

5. **Answer:** A) Uniform acceleration

Solution: When the velocity of a body changes by strictly equal amounts in regular, equal intervals of time, the rate of change is constant. This meets the exact definition of uniform acceleration.

6. **Answer:** B) 27.27 km/h

Solution: The average speed for a round trip with speeds $v_1 = 30$ km/h and $v_2 = 25$ km/h covering equal distances is given by the formula Average Speed $= \frac{2v_1v_2}{v_1+v_2}$. The calculation is $\frac{2\times30\times25}{30+25} = \frac{1500}{55}$, which equals approximately 27.27 km/h.

7. **Answer:** A) $3m/s^2$

Solution: The scooter starts from rest ($u = 0$) and acquires a velocity of 54 km/h. Converting to m/s: $54 \times \frac{5}{18} = 15$ m/s. The time is 5 s. Using the formula $a = \frac{v-u}{t}$, the calculation is $\frac{15-0}{5} = 3$ m/s^2.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

8. **Answer:** B) Non-uniform acceleration

Solution: If a body moves with a changing (decreasing) acceleration, its rate of change of velocity is not remaining constant. Any acceleration that is not strictly constant over time falls under the general category of non-uniform acceleration.

9. **Answer:** A) The velocity is a vector quantity & B) The rate of displacement is called velocity

Solution: Velocity is defined by both speed and direction, making it a vector quantity. Furthermore, velocity is mathematically defined as the rate of change of displacement with respect to time.

10. **Answer:** A) Velocity is a vector, B) SI unit of velocity is m/s, & C) If displacement = 0 $\vec{v} = 0$

Solution: Velocity requires both magnitude and direction, classifying it as a vector. The standard SI unit for velocity is meters per second or m/s. Average velocity is total displacement over time, so if displacement is zero, the average velocity equates to zero.

11. **Answer:** B) Less than or equal to one

Solution: Average velocity is calculated based on displacement, while average speed is calculated based on distance. Because displacement is defined as the shortest path between two points, it can never exceed the total distance traveled. Therefore, the ratio of average velocity to average speed must always be less than or equal to one.

12. **Answer:** A, B,
Solution: Acceleration is defined as the rate of change of velocity. Since velocity is a vector, an acceleration is produced if its magnitude (speed) changes, its direction changes, or both change simultaneously.

13. **Answer:** A) Displacement, B) Velocity, & C) Acceleration
Solution: Vector quantities require both a magnitude and a specific direction. Displacement, velocity, and acceleration all possess direction inherently, classifying them as vectors. Distance only has magnitude, so it is a scalar.

Passage – I

14. **Answer:** B) 750 m
Solution: Ahmed's velocity is 45 km/h. Converting this to m/s: $45 \times \frac{5}{18} = 12.5$ m/s. The time given is one minute, which equals 60 seconds. Using the formula Distance = Speed × Time, the calculation is $12.5 \times 60 = 750$ m.

15. **Answer :** A) 12.5 m
Solution : Following the previous conversion, Ahmed's speed is 12.5 m/s. The time given is one second. Using the formula Distance = Speed × Time, the calculation is $12.5 \times 1 = 12.5$ m.

Passage – II

16. **Answer:** C) 324 km-south
Solution: Displacement is a vector quantity that represents the shortest distance from the initial to the final position along with its direction. The aeroplane travels 324 km strictly towards the south. Therefore, its total displacement is 324 km south.

17. **Answer:** A) 972
Solution: The displacement is 324 km and the time taken is 20 minutes. To find velocity in km/hr, time must be converted to hours: 20 minutes $= \frac{20}{60}$ hours $= \frac{1}{3}$ hours. Using the formula Velocity $= \frac{\text{Displacement}}{\text{Time}}$, the calculation is $\frac{324}{\frac{1}{3}} = 324 \times 3 =$ 972 km/hr.

18. **Answer:** C) 60 km/hr
Solution: The aeroplane covers a displacement of 90 km towards the east in 1.5 hours (written as $1\frac{1}{2}$ hours). Using the formula Velocity $= \frac{\text{Displacement}}{\text{Time}}$, the calculation is $\frac{90}{1.5} = 60$ km/hr.

ASSERTION AND REASON TYPE

19. **Answer:** B) A is false, but R is true
Solution: Assertion A claims that non-uniform speed covers equal distances in equal intervals of time, which is actually the exact definition of uniform speed; thus, A is false. Reason R states that in non-uniform motion, speed changes with time, which is functionally true. Therefore, A is false and R is true.

20. **Answer:** B) Both A and R are true, but R is not the correct explanation of A
Solution: Assertion A correctly defines instantaneous speed as the speed measured at a particular instant in time. Reason R correctly states that a speedometer displays instantaneous speed. Both statements are true independently, but a speedometer's mechanical function does not inherently serve as the physical explanation for the definition of instantaneous speed.

MATRIX MATCHING TYPE

21. **Answer:** a-Q, b-S, c-R, d-T

Solution: Based on standard definitions: a) 1 km/hr directly converts to $\frac{5}{18}$ m/s (Q) .b) A speedometer is an instrument that measures instantaneous speed (S) . c) Average velocity is calculated using the formula Total displacement divided by Total time taken (R) . d) Uniform speed is defined as covering equal distances in equal intervals of time (T).

22. **Answer:** Speed - Distance travelled per unit time, Velocity - Rate of change of displacement, Acceleration - Rate of change of velocity, Average Speed - Total distance / total time

Solution: By standard kinematics definitions provided in the text: Speed is the distance traveled per unit time. Velocity is the vector equivalent, representing the rate of change of displacement. Acceleration tracks the rate of change of velocity. Lastly, average speed is calculated as the total path distance divided by the total time taken.

INTEGER TYPE

23. **Answer:** Average speed: 3 m/s, Average velocity: 0 m/s

Solution: Usha swims a total distance of 180 m (back and forth in a 90 m pool) in 1 minute (60 s). Using the formula Average Speed $= \frac{\text{Total Distance}}{\text{Total Time}}$, the calculation is $\frac{180}{60} = 3$ m/s. Since her final position is identical to her initial position after the round trip, her displacement is zero. Using Average Velocity $= \frac{\text{Displacement}}{\text{Time}}$, the calculation is $\frac{0}{60} = 0$ m/s.

24. **Answer:** 5

Solution: The body travels a distance of 10 m in a time of 2 s. Using the formula Speed $= \frac{\text{Distance}}{\text{Time}}$, the calculation is $\frac{10}{2} = 5$ m/s.

25. **Answer:** 2.5

Solution: The pot is 10 m away, making one full trip (there and back) equal to 20 m. For 5 trips, the total distance is $5 \times 20 = 100$ m. The total time taken is 40 seconds. Using the formula Average Speed $= \frac{\text{Total Distance}}{\text{Total Time}}$, the calculation is $\frac{100}{40} = 2.5$ m/s.

3. Force And Laws Of Motion

LEVEL-I

1. **Answer :** B) Mass
Inertia is a property of matter that depends entirely on the mass of the object.
Heavier objects have more inertia.

2. **Answer :** D) Acceleration
According to Newton's second law, force is the product of mass and acceleration.
$F = m \cdot a$
A constant force acting on a constant mass will produce a uniform acceleration.

3. **Answer :** C) Inertia
The tendency of the body to continue in its state of rest or uniform motion in a straight line, even when some external force acts on it, is defined as inertia.

4. **Answer :** C) Inertia of direction
The sparks fly off tangentially because they tend to maintain their original direction of motion.
This happens due to the inertia of direction.

5. **Answer :** C) Both have same inertia
Inertia depends exclusively on the mass of the objects.
Since both the iron and wood spheres have equal mass, they will have the exact same inertia.

6. **Answer :** B) Weight
The gravitational force of attraction of the earth acting on a body is known as the weight of that body.

7. **Answer :** A) Vertically downwards
Gravity pulls all objects toward the center of the earth.
Therefore, the weight of an object is always directed vertically downwards.

8. **Answer :** C) Inertia
Newton's first law of motion explains the concept of inertia.
It states that a body will resist any change to its state of rest or motion.

9. **Answer :** B) Inertia of motion
The fan blades are initially in a continuous state of motion.
When switched off, they continue to move for some time due to the inertia of motion.

10. **Answer :** B) 4 kg
Inertia is directly proportional to mass.
Since 4 kg > 1 kg, the 4 kg body will have more inertia.

11. **Answer :** C) Both A and B
According to Newton's first law of motion, unless some external force acts, a body will continue in its state of rest or its state of motion in a straight line.

12. **Answer :** B) Move forward
When a running car stops suddenly, the lower part of the passengers' bodies comes to rest.
The upper part of their bodies continues moving forward due to the inertia of motion, causing a forward jerk.

13. **Answer :** C) Force
The physical cause which changes or tends to change the state of rest or the state of motion of a body is defined as force.

14. **Answer :** A) Law of inertia
Newton's first law defines the property of inertia.
Hence, it is also called the law of inertia.

15. **Answer :** C) Inertia of rest keeps the upper part of body at rest whereas lower part moves forward with the horse
When the horse suddenly starts running, the lower part of the rider's body moves forward with the horse.
The upper part of the body tends to remain in its initial state due to the inertia of rest, causing the rider to fall back.

16. **Answer :** C) Upper part of the body continues to be in state of motion whereas lower part of the body in contact. With seat remain at rest
When a train stops suddenly, the seat and the lower part of the body remain at rest.
The upper part of the body continues to be in a state of motion due to the inertia of motion, causing an instant forward jerk.

17. **Answer :** D) Unable to change by itself the state of rest and uniform linear motion
Inertia is the property by virtue of which a body is unable to change by itself its state of rest or uniform motion.

18. **Answer :** B) Due to inertia of motion upper part of the body continues to be in motion in the forward direction. While feet comes to rest
When getting down from a running bus, the feet immediately come to rest upon touching the ground.
The upper part of the body continues its forward motion due to the inertia of motion, causing the man to fall forward.

LEVEL-II

SINGLE CORRECT OPTION TYPE

1. **Answer :** D) Exactly in the hand which threw it up
The ball shares the horizontal velocity of the moving car.
Due to the inertia of motion, the ball continues to move forward at the same constant horizontal velocity as the car, landing exactly back in the hand.

2. **Answer :** A) Inertia of direction
Passengers in a moving bus tend to travel in a straight line.
When the bus suddenly takes a turn, they are thrown outward because they naturally resist the change in their direction of motion due to the inertia of direction.

3. **Answer :** D) By running he gives himself large inertia of motion
An athlete runs some distance before taking a long jump to build up forward velocity.
This provides a large inertia of motion, which helps carry the athlete over a longer distance during the jump.

4. **Answer :** C) Unbalanced
A net external force is required to change a body's state of rest or uniform motion.
Only unbalanced forces can move a stationary body or stop a moving body.

5. **Answer :** C) The combination of forces acting on it balance each other
When a body is stationary, it means the net force acting on it is exactly zero.
This implies that the combination of all forces acting on the body balance each other out.

6. **Answer :** B) Inertia of rider
When the horse suddenly starts moving, the lower half of the rider moves forward.
The upper half resists this change due to the inertia of rest, causing the rider to fall backwards.

7. **Answer :** C) Inertia of rest
When a car or bus starts suddenly, the lower part of the body moves with the vehicle.
The upper part of the body remains in its original position due to the inertia of rest, causing passengers to fall backward.

8. **Answer :** A) Inertia of motion
When a running car or bus stops suddenly, the lower part of the body comes to a halt.
The upper part of the body continues moving forward due to the inertia of motion, causing passengers to jerk forward.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. **Answer :** A, B) Weight, Tension
While pulling water in a bucket from a well, gravity pulls the bucket downwards, which is its weight.
The rope exerts an upward pull to lift the bucket, which is the tension force.

10. **Answer :** A, B, C, D) Stationary object into motion, Moving object to rest, Change direction of moving body, Change dimensions of an object
A force is capable of moving a stationary object and stopping a moving object.
It can also change the direction of a moving body and change the dimensions (shape and size) of an object.

11. **Answer :** A, B, C, D) Muscular force, Mechanical force, Electrostatic force, Magnetic force
Forces can be categorized into contact forces, such as muscular and mechanical forces.
They can also be non-contact forces, such as electrostatic and magnetic forces.

12. **Answer :** C, D) Mass is a measure of inertia, It explains the quality of force
Newton's first law introduces the concept of inertia, establishing that mass is a quantitative measure of inertia.
It also provides a qualitative definition of force as an external push or pull that changes a body's state of motion.

ASSERTION AND REASON TYPE

13. **Answer :** A) Both A and R are true and R is the correct explanation of A
Balanced forces result in a net force of zero acting on a body.
Since the resultant is zero, there is no acceleration, meaning these forces cannot move a stationary body or stop a moving body.

14. **Answer :** A) Both A and R are true and R is the correct explanation of A
When a bus starts suddenly, passengers fall backwards.
This occurs because the lower part of the body moves with the bus, while the upper part tends to remain at rest due to the inertia of rest, making the reason the correct explanation.

ASSERTION AND REASON TYPE

15. **Answer :** A) Inertia of rest
A person in a car tends to fall back when it suddenly starts.

This happens due to the upper body's tendency to resist the sudden forward motion, which is the inertia of rest.

16. **Answer :** C) Inertia of direction
When the string is cut, the inward centripetal force is removed.
The stone flies off tangentially because it continues in its instantaneous direction of motion due to the inertia of direction.

17. **Answer :** B) Inertia of motion
When the moving bus stops suddenly, the passenger's lower body comes to rest.
The upper body leans forward as it continues to remain in its state of motion due to the inertia of motion.

18. **Answer :** B) Inertia of motion of the boxes
The loaded truck and the boxes were moving at a constant speed.
When brakes are applied, the boxes shift forward because they tend to maintain their previous state of forward motion due to the inertia of motion.

19. **Answer :** C) The truck experiences a retarding force while boxes tend to maintain their previous state of motion
Braking applies a retarding force to the truck, slowing it down.
The loosely placed boxes do not experience enough immediate opposing force, so they tend to maintain their previous state of forward motion.

20. **Answer :** C) Change shape and size of an object
The loosely tied ropes stretched and changed their shape due to the forces acting on them during braking.
This directly demonstrates that force can change the shape and size of an object.

ASSERTION AND REASON TYPE

21. **Answer :** a → T, b → P and S, c → R, d → Q
Muscular force corresponds to pushing a lawn roller by physical effort.
Mechanical force corresponds to forces generated by machines, such as running a motor car engine or a steam engine.
Gravitational force corresponds to the earth's pull, which makes a stone fall downwards.
Electrostatic force corresponds to the attraction of small pieces of paper by a charged comb.

Momentum and Newton's Laws of Motion

LEVEL-1

1. **Answer :** A) Force
$F = m \times a$

2. **Answer :** A) 9.8
1 kgf = 9.8 N

3. **Answer :** B) Momentum
$p = m \times v$

4. **Answer :** C) Linear momentum
Linear momentum is the measure of the total quantity of motion contained in a body.

5. **Answer :** B) p
Linear momentum is universally denoted by the letter p.

6. **Answer :** B) Momentum
1 Ns = 1 kg $m/s^2 \times$ s
1 Ns = 1 kg m/s

This is the standard unit of momentum.

7. **Answer :** A) 1 $kg\,m/s$

$m = 200$ g $= 0.2$ kg

$v = 5$ m/s

$p = m \times v$

$p = 0.2 \times 5$

$p = 1$ kg m/s

8. **Answer :** B) Velocity must be reduced to half

For a given body, its mass remains constant.

$p = m \times v$

To halve the momentum, the velocity must be halved.

9. **Answer :** D) 1:3

$p_1 = m \times v_1 = m \times v$

$p_2 = m \times v_2 = m \times 3v$

$\frac{p_1}{p_2} = \frac{mv}{3mv}$

$\frac{p_1}{p_2} = \frac{1}{3}$

10. **Answer :** B) Lorry comes to rest in shorter distance

$KE = \frac{p^2}{2m}$

Since p is the same, the body with larger mass (lorry) has less kinetic energy.

$F \times s = KE$

With the same braking force F, smaller KE results in a smaller stopping distance s.

11. **Answer :** B) Its momentum is doubled

$p = m \times v$

If velocity v is doubled, momentum p is directly doubled.

12. **Answer :** B) 8 m/s

$a = \frac{F}{m}$

$a = \frac{15}{15}$

$a = 1$ m/s^2

$v = u + a \times t$

$v = 0 + 1 \times 8$

$v = 8$ m/s

13. **Answer :** C) Magnitude of force

Newton's second law provides the mathematical formula $F = m \times a$ to quantify and measure force.

14. **Answer :** B) An acceleration of 1 ms^{-2} on a body of mass 1 kg

1 N = 1 kg × 1 m/s^2

15. **Answer :** B) $1N = 10^5$ Dyne

1 N = 1 kg × 1 m/s^2

1 N = 1000 g × 100 cm/s^2

1 N = 10^5 Dyne

16. **Answer :** A) Mass

$p = m \times v$

Momentum is directly proportional to both mass and velocity.

17. **Answer :** B) The ground on the horse
The horse pushes the ground backward (action).
The ground pushes the horse forward (reaction).

18. **Answer :** A) Backward
The person pushes the boat backward (action) to jump forward (reaction).

19. **Answer :** C) Equal, Opposite
Newton's third law states that for every action, there is an equal and opposite reaction.

20. **Answer :** A) Action
The force exerted by the first body on the second is termed the action force.

21. **Answer :** B) Never cancel
Action and reaction forces act on two different bodies, so they cannot cancel each other.

22. **Answer :** A) 10 N
Action = Reaction
Reaction = 10 N

23. **Answer :** C) Newton's 3rd law
The backward recoil of the gun is the equal and opposite reaction to the forward action of the fired bullet.

24. **Answer :** B) 25 $kg\ ms^{-1}$
$m = 5$ kg
$v = 5$ m/s
$p = m \times v$
$p = 5 \times 5$
$p = 25$ kg m/s

25. **Answer :** A) 2 kg
$F = 20$ N
$a = 10$ m/s^2
$m = \frac{F}{a}$
$m = \frac{20}{10}$
$m = 2$ kg

26. **Answer :** D) 12 kg
$F = 24$ N
$a = 2$ m/s^2
$m = \frac{F}{a}$
$m = \frac{24}{2}$
$m = 12$ kg

LEVEL-II

SINGLE OPTION CORRECT TYPE

1. **Answer :** D) All the above
A force can change the state of motion (speed or direction) and the shape of an object.

2. **Answer :** B) Measure of force
Newton's second law provides the quantitative equation $F = m \times a$.

3. **Answer :** C) Resultant force on it is zero
According to $F = m \times a$, acceleration a will be zero only when the net resultant force F becomes zero.

4. **Answer :** B) Action = -reaction
The negative sign indicates the opposite direction of the reaction force.

5. **Answer :** A) 48 $kg\ ms^{-1}$
$F = \frac{\Delta p}{t}$
$\Delta p = F \times t$
$\Delta p = 12 \times 4$
$\Delta p = 48$ kg m/s

6. **Answer :** C) $10^{-3}N$
$m = 20$ g $= 0.02$ kg
$a = 5$ cm/s^2 $= 0.05$ m/s^2
$F = m \times a$
$F = 0.02 \times 0.05$
$F = 0.001$ N
$F = 10^{-3}$ N

7. **Answer :** B) The different bodies
Action and reaction forces always exist in pairs and act simultaneously on two different interacting bodies.

8. **Answer :** C) Newton's third law of motion
The recoil of a cannon is the reaction force to the action of the projectile being fired forward.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. **Answer :** A, B, C, D
All of these actions rely on the principle of equal and opposite reaction to produce forward motion.

10. **Answer :** A) 8 $kg\ m/s$
$m = 2$ kg
$\Delta v = 8 - 4 = 4$ m/s
$\Delta p = m \times \Delta v$
$\Delta p = 2 \times 4$
$\Delta p = 8$ kg m/s

11. **Answer :** A, B
Linear momentum is a measure of the quantity of motion, and impulse is precisely defined as the change in momentum.

12. **Answer :** A, C
$a = \frac{v - u}{t}$
$a = \frac{6 - 5}{2}$
$a = 0.5$ m/s^2
$m = \frac{F}{a}$
$m = \frac{0.6}{0.5}$
$m = 1.2$ kg
1.2 kg = 1200 g

13. **Answer :** A, B, C
By Newton's second law, the rate of change of momentum is proportional to the net applied force and occurs in its direction; in standard SI units, the constant of proportionality is 1, making them equal.

ASSERTION AND REASON TYPE

14. **Answer :** A) Both A and R are true and R is the correct explanation of A
A ball with higher speed has more momentum ($p = m \times v$) and transfers a larger impulse upon impact, causing more hurt.

15. **Answer :** A) Both A and R are true and R is the correct explanation of A
$a = \frac{F}{m}$
For a constant force, acceleration is inversely proportional to mass, so the lighter body accelerates more.

COMPREHENSION TYPE

16. **Answer :** A) Action
The initial push exerted by the man is considered the action force.

17. **Answer :** A) 5N
Reaction = Action
Reaction = 5N

18. **Answer :** A) 5N
The force applied is 5N.

19. **Answer :** C) 30
$p_{initial} = m \times u$
$p_{initial} = 3 \times 10$
$p_{initial} = 30$ kg m/s

20. **Answer :** D) 60
$p_{final} = m \times v$
$p_{final} = 3 \times 20$
$p_{final} = 60$ kg m/s

21. **Answer :** B) 30
$\Delta p = p_{final} - p_{initial}$
$\Delta p = 60 - 30$
$\Delta p = 30$ kg m/s

MATRIX MATCHING TYPE

22. **Answer :** a → Q, b → S, c → P, d → R, T
Force is measured in kg m/s^2 (Q).
Newton's 1st law describes Inertia (S).
Newton's 2nd law explains force quantitatively (P).
Momentum is mass in motion (R) and its unit is kg m/s (T).

23. **Answer :** a → S, b → Q, c → R, d → P, T
Weight is defined as $m \times g$ (S).
Action equals negative Reaction (Q).
$mv - mu$ represents the change in momentum (R).
Newton's 3rd law applies to Rockets (P) and Rowing (T).

INTEGER TYPE

24. **Answer :** B) $1ms^{-2}$
"2 kg" of force typically implies 2 kgf in this context.
$F = 2$ kgf $= 2 \times 10 = 20$ N

$a = \frac{F}{m}$

$a = \frac{20}{20}$

$a = 1 \text{ m/s}^2$

25. **Answer :** B) 75 $kg\ m/s$

$p = m \times v$

$p = 15 \times 5$

$p = 75 \text{ kg m/s}$

26. **Answer :** A) 100 $kg.m/s$

$p = m \times v$

$p = 10 \times 10$

$p = 100 \text{ kg m/s}$

Previous Year Questions

1. **Answer :** A) (Option showing wind vector right, gravity down, string tension up-left)
Gravity pulls the bob straight down.
The wind blows horizontally, pushing the bob to the right.
The tension in the string pulls back up and to the left toward the pivot point.

2. **Answer :** D) (i), (ii) and (iii)

$F_{crew} = m_{crew} \times (g + a)$

$F_{crew} = 500 \times (10 + 15)$

$F_{crew} = 12500 \text{ N} = 1.25 \times 10^4 \text{ N}$ downwards

$F_{rotor} = (m_{heli} + m_{crew}) \times (g + a)$

$F_{rotor} = 2500 \times (10 + 15)$

$F_{rotor} = 62500 \text{ N} = 6.25 \times 10^4 \text{ N}$ downwards

The air pushes back up on the helicopter with an equal reaction force of 6.25×10^4 N.

3. **Answer :** B) BC
Force is zero when acceleration is zero.
Zero acceleration means constant velocity, which is represented by a straight line with a constant slope on a displacement-time graph (region BC).

4. **Answer :** C) P - (iv), Q- (iii), R- (ii), S- (i)
Newton's 1st law provides the qualitative definition of force.
Newton's 2nd law provides the quantitative measure of force.
Impulse is the effect of force over time.
Recoiling of a gun is an application of Newton's 3rd law.

5. **Answer :** A) 2Mu

$p_{initial} = M \times u$

$p_{final} = M \times (-u)$

$\Delta p = |p_{final} - p_{initial}|$

$\Delta p = |-Mu - Mu|$

$\Delta p = 2Mu$

4. Work, Energy And Power

Work

LEVEL-1

1. **Answer :** A) Force × distance moved in the direction of applied force
Work done is defined as the product of the force applied and the distance moved in the direction of the force.

2. **Answer :** C) When a force produces motion
Work is only said to be done if a force displaces a body.

3. **Answer :** A) Positive
The agent applies a force in the direction of motion to overcome the electrostatic repulsion, so the work done by the agent is positive.

4. **Answer :** B) Negative
Work done is negative when a force acts opposite to the direction of motion of the body.

5. **Answer :** C) Zero
In a circular path, the centripetal force is always perpendicular to the direction of displacement, so the work done is zero.

6. **Answer :** A) Positive
When the displacement is in the same direction as the applied force, the work done is positive.

7. **Answer :** C) Zero
No work is done by the force if it fails to displace the body.

8. **Answer :** B) Negative
When brakes are applied on a moving vehicle, the work done by the braking force is negative because the force is opposite to the displacement.

9. **Answer :** A) Negative because the force and displacement are in opposite direction
The gravitational force acts downwards, but the bucket is drawn upwards.
Since the force and displacement are in opposite directions, the work done is negative.

10. **Answer :** B) newton, metre
1 joule is the work done when a force of one newton moves an object through a distance of one metre.

11. **Answer :** D) Zero
While holding a suitcase, there is no displacement.
Because the displacement is zero, the work done is zero.

12. **Answer :** B) Force, displacement
Work is calculated as the product of the force applied and the displacement of the object.

13. **Answer :** D) 4900 J
$W = m \cdot g \cdot h$
$W = 100 \cdot 9.8 \cdot 5$
$W = 4900$ J

14. **Answer :** A) Zero
The force of gravity acts vertically downwards, which is perpendicular to the horizontal displacement.
When a body moves perpendicular to the force, no work is done.

15. **Answer :** B) Scalar
Energy and work are both scalar quantities because they only have magnitude and no direction.

16. **Answer :** B) mgh
The work done in raising a body of mass m to a height h is equal to its potential energy.
$W = m \cdot g \cdot h$

17. **Answer :** D) -10000 J
The formula for work done by gravity is:
$W = -m \cdot g \cdot h$
Assuming $g = 10\ \text{m/s}^2$:
$W = -100 \cdot 10 \cdot 10$
$W = -10000\ \text{J}$

18. **Answer :** A) 50000
Assuming $g = 10\ \text{m/s}^2$:
$W = m \cdot g \cdot h$
$W = 50 \cdot 10 \cdot 100$
$W = 50000\ \text{J}$

19. **Answer :** A) 196 J
First, convert the distance to meters:
$h = 400\ \text{cm} = 4\ \text{m}$
$W = m \cdot g \cdot h$
$W = 5 \cdot 9.8 \cdot 4$
$W = 196\ \text{J}$

20. **Answer :** D) 4J
$W = F \cdot s$
$W = 1 \cdot 4$
$W = 4\ \text{J}$

21. **Answer :** B) $5 \times 10^8 J$
First, convert the distance to meters:
$s = 1\ \text{km} = 1000\ \text{m}$
The engine must apply a force equal to the frictional resistance to pull the train:
$W = F \cdot s$
$W = 5 \times 10^5 \cdot 1000$
$W = 5 \times 10^8\ \text{J}$

LEVEL-II

SINGLE OPTION CORRECT TYPE

1. **Answer :** A) 196 J
Convert distance to meters:
$h = 400\ \text{cm} = 4\ \text{m}$
$W = m \cdot g \cdot h$
$W = 5 \cdot 9.8 \cdot 4$
$W = 196\ \text{J}$

2. **Answer :** C) 0J
The girl applies an upward force to hold the bag, but moves horizontally.
Since the force and displacement are perpendicular, the work done against gravity is zero.

3. **Answer :** B) 800 J
Weight is the force $F = 400$ N.
$W = F \cdot h$
$W = 400 \cdot 2$
$W = 800$ J

4. **Answer :** C) 40 J
First, calculate the horizontal force applied to accelerate the block:
$F = m \cdot a$
$F = 2 \cdot 0.5$
$F = 1$ N
Calculate the work done by this horizontal force:
$W = F \cdot s$
$W = 1 \cdot 40$
$W = 40$ J

5. **Answer :** B) 400 J
$W = -F \cdot s$
$W = -100 \cdot 4.0$
$W = -400$ J
Magnitude = 400 J

6. **Answer :** C) Zero
The weight of the body acts vertically downwards while the displacement is horizontal.
Because the force is perpendicular to the displacement, the work done is zero.

7. **Answer :** C) 50 kg
Assume $g = 10$ m/s^2:
$W = m \cdot g \cdot h$
$2500 = m \cdot 10 \cdot 5$
$m = \frac{2500}{50}$
$m = 50$ kg

8. **Answer :** B) 121 N
Convert displacement to meters:
$s = 20$ cm $= 0.2$ m
$W = F \cdot s$
$24.2 = F \cdot 0.2$
$F = \frac{24.2}{0.2}$
$F = 121$ N

9. **Answer :** C) The initial velocity of the body
Work done relies on force, displacement, and the angle between them.
It does not depend on the initial velocity of the body.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

10. **Answer :** A, C
The work done depends directly on the magnitude of the applied force and the distance through which the body moves.

11. **Answer :** A, C
When displacement is opposite to the applied force, the work done is negative, calculated as $W = -(F \cdot s)$.
This explains why the work done by a frictional force is negative.

12. **Answer :** A, B, D
When whirled in a circle, tension is perpendicular to velocity, so work is zero.
When brakes are applied, the braking force is opposite to motion, so work is negative.
A flying aeroplane is propelled by the engine thrust, which acts approximately in the direction of motion. Hence, the engine does positive work.
freely falling body has displacement and gravity in the same direction, so work is positive.

13. **Answer :** A, C, D
Work is zero if the force is perpendicular to velocity, if the object is stationary, or if the specific point of application of the force does not move.

ASSERTION AND REASON TYPE

14. **Answer :** A) Both A and R are true and R is the correct explanation of A
When a body moves perpendicular to the force, the displacement along the direction of the force is zero.
Because work is the product of force and displacement along its direction, the resulting work done is zero.

15. **Answer :** B) Both A and R are true but R is not the correct explanation of A
Gravity acts downwards while the object is lifted upwards, making their directions opposite, which results in negative work.
Reason (R) states the condition for positive work, which is factually true, but it does not directly explain why opposite directions yield negative work.

COMPREHENSION TYPE

16. **Answer :** A) 10 J
$W = F \cdot s$
$W = 10 \cdot 1$
$W = 10$ J

17. **Answer :** C) 12000 J
$W = m \cdot g \cdot h$
$W = 200 \cdot 10 \cdot 6$
$W = 12000$ J

18. **Answer :** C) 225 J
Assume $g = 10$ m/s^2:
$W = m \cdot g \cdot h$
$W = 15 \cdot 10 \cdot 1.5$
$W = 225$ J

19. **Answer :** B) Zero
When the man is stationary, the displacement is zero.
Therefore, the work done is zero.

20. **Answer :** B) Zero
When the man moves horizontally, the upward force he exerts to support the load is perpendicular to the horizontal displacement.
Therefore, no work is done against gravity.

21. **Answer :** A) 20580 J
To move vertically, the man must lift both his own mass and the supported mass.
$m_{total} = 50 + 20 = 70$ kg
Assuming $g = 9.8$ m/s^2:
$W = m_{total} \cdot g \cdot h$

$W = 70 \cdot 9.8 \cdot 30$

$W = 20580$ J

MATRIX MATCHING TYPE

22. **Answer :** a → S or T, b → R, c → Q or R, d → P, e → P

Work done by a force displacing an object in its own direction is $F \cdot s$ (S) is positive (T).

Work done by gravity on an upward-thrown object is negative (R).

Work done by frictional force is negative, represented as $-F \cdot s$ (Q) is negative (R).

Work done perpendicular to the direction of force is zero (P).

Work done when unable to move a rock is zero (P).

23. **Answer :** a → P, Q, b → T, c → S, d → R

1 Joule equals 10^7 erg (P) and its SI base units are 1 kg m^2s^{-2} (Q).

1 erg in CGS units is 1 g cm^2s^{-2} (T).

Work done in lifting a body is the product of weight and vertical distance (S).

A body falling freely under gravity experiences positive work done by the gravitational force (R).

INTEGER TYPE

24. **Answer :** 35280 J or 36000 J

Assume $g = 9.8$ m/s^2:

$W = m \cdot g \cdot h$

$W = 50 \cdot 9.8 \cdot 72$

$W = 35280$ J

(Using $g = 10$ m/s^2 yields 36000 J).

25. **Answer :** 0.2 J

Convert the displacement to meters:

$s = 10$ cm $= 0.1$ m

$W = F \cdot s$

$W = 2 \cdot 0.1$

$W = 0.2$ J

26. **Answer :** 3920 J

$W = m \cdot g \cdot h$

$W = 200 \cdot 9.8 \cdot 2$

$W = 3920$ J

27. **Answer :** 0 J

The coolie exerts an upward vertical force to carry the load, but walks horizontally. Because the displacement is perpendicular to the force applied, the work done is zero.

Energy and Power

LEVEL-I

1. **Answer :** C) Energy

The capacity or ability to do work is defined as energy.

2. **Answer :** A) 1 m

By definition, one joule is the amount of work done when a force of 1 N displaces a body by 1 m in the direction of the force.

3. **Answer :** B) Scalar

Energy only has magnitude and no direction, making it a scalar quantity.

4. **Answer :** B) Joule

The Standard International (SI) unit of energy is the Joule.

5. **Answer :** D) Both B and C
Potential energy is the energy possessed by a body due to its position or deformed shape.
Both a compressed spring and a stretched elastic band possess elastic potential energy due to their deformed shapes.

6. **Answer :** D) Chemical
Fuels store chemical energy.
When they burn, this chemical energy is released and converted into heat energy.

7. **Answer :** D) Potential
Water stored behind a dam at a height possesses potential energy due to its elevated position in the earth's gravitational field.

8. **Answer :** A) P
Gravitational potential energy depends on height ($m \cdot g \cdot h$).
Point P is at the highest elevation, so the ball has the maximum potential energy at this point.

9. **Answer :** D) The kinetic energy from the stretched elastic band was changed into kinetic energy and transferred from the marble to the ball

10. **Answer :** B) Kinetic
When a substance is heated, its thermal energy increases.
This causes the molecules to move and vibrate faster, meaning their kinetic energy increases.

11. **Answer :** A) A only
The stretched spring/string (part A) of the bow stores elastic potential energy when pulled back.
This stored energy is converted into the kinetic energy of the arrow upon release.

12. **Answer :** A) A
Potential energy is directly proportional to height.
Point A is the lowest point in the trajectory, meaning it has the least potential energy.

13. **Answer :** B) Point 2
At point 1 (top), the ball has only potential energy.
At point 3 (just before hitting the ground), it has maximum kinetic energy and zero potential energy.
At point 2 (mid-air), it possesses both potential and kinetic energy as it falls.

14. **Answer :** D) Both A & C
The marble stops because the frictional force between the marble and the floor opposes its motion.
This friction converts the marble's kinetic energy into heat and sound energy.

15. **Answer :** B) Chemical energy
During photosynthesis, plants convert light energy from the sun into chemical energy stored in food.

16. **Answer :** D) 2 minutes
$P = 2000$ W
$E = 240000$ J
$P = \frac{E}{t}$
$t = \frac{E}{P}$

$t = \frac{240000}{2000}$

$t = 120$ s

$t = 2$ minutes

17. **Answer :** D) 6 w

$W = 30$ J

$t = 5$ s

$P = \frac{W}{t}$

$P = \frac{30}{5}$

$P = 6$ W

18. **Answer :** C) Unit of power is joule

This statement is incorrect because the unit of power is the Watt (or Joules per second), whereas the Joule is the unit of work and energy.

19. **Answer :** B) 10 sec

$P = 100$ W

$E = 1000$ J

$t = \frac{E}{P}$

$t = \frac{1000}{100}$

$t = 10$ s

20. **Answer :** C) Watt

The SI unit of power is the Watt, which is equivalent to one Joule per second.

21. **Answer :** A) 25 w

$F = 600$ N

$s = 20$ m

$W = F \cdot s$

$W = 600 \cdot 20$

$W = 12000$ J

$t = 8$ minutes $= 480$ s

$P = \frac{W}{t}$

$P = \frac{12000}{480}$

$P = 25$ W

22. **Answer :** C) Power

Work done in relevance to time (the rate of doing work) is defined as power.

23. **Answer :** B) 50 w

$E = 90000$ J

$t = 30$ minutes $= 1800$ s

$P = \frac{E}{t}$

$P = \frac{90000}{1800}$

$P = 50$ W

24. **Answer :** A) $1 J/s$

One watt is defined as the rate of one joule of work done per second.

25. **Answer :** D) Energy
Kilowatt-hour (KWH) is the commercial unit of electrical energy.

26. **Answer :** C) 200 W
$m = 60$ kg
$h = 10$ m
$g = 10$ m/s^2
$W = m \cdot g \cdot h$
$W = 60 \cdot 10 \cdot 10$
$W = 6000$ J
$t = 30$ s
$P = \frac{W}{t}$
$P = \frac{6000}{30}$
$P = 200$ W

27. **Answer :** B) $\frac{1\,mv^2}{2\,t}$
$W = \Delta K.E$
$W = \frac{1}{2} \cdot m \cdot v^2 - 0$
$W = \frac{1}{2} \cdot m \cdot v^2$
$P = \frac{W}{t}$
$P = \frac{\frac{1}{2} \cdot m \cdot v^2}{t}$
$P = \frac{m \cdot v^2}{2 \cdot t}$

LEVEL-II

SINGLE OPTION CORRECT TYPE

1. **Answer :** B) Kinetic energy is converted into sound and thermal energy
When the car applies brakes to stop, its kinetic energy is lost to friction, which converts it into thermal (heat) energy and sound energy.

2. **Answer :** A) Converted or) transferred
According to the law of conservation of energy, energy cannot be created or destroyed, but it can only be converted from one form to another or transferred between objects.

3. **Answer :** C) Gravitational potential energy
The energy possessed by an object by virtue of its elevated height against the earth's gravity is called gravitational potential energy.

4. **Answer :** B) 9J
$m = 2$ kg
$v = 3$ m/s
$E_k = \frac{1}{2} \cdot m \cdot v^2$
$E_k = \frac{1}{2} \cdot 2 \cdot 3^2$
$E_k = 9$ J

5. **Answer :** B) Scalar
Power is the rate of doing work.

Since work and time are both scalar quantities, power is also a scalar quantity.

6. **Answer :** B) 746 W

One Horsepower (HP) is a practical unit of power equal to 746 Watts.

7. **Answer :** B) $\frac{W}{T}$

Power is mathematically defined as the ratio of work done (W) to the time taken (T).

$P = \frac{W}{T}$

8. **Answer :** B) 1 W

$W = 15$ J

$T = 15$ s

$P = \frac{W}{T}$

$P = \frac{15}{15}$

$P = 1$ W

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. **Answer :** B, D

If the kinetic energy increases, the velocity magnitude increases.

This means the work done by the net force is positive, so the angle between the force and velocity is less than 90°.

Also, since velocity increases, the magnitude of linear momentum ($p = m \cdot v$) increases continuously.

10. **Answer :** B, C, D

A force can do positive work (increasing K.E), negative work (decreasing K.E), or stop the body completely (making K.E zero).

11. **Answer :** A, B, C, D

All statements are correct characteristics and formulas for power.

Power is work done per unit time, is inversely proportional to time, has the dimensional formula $[M^1L^2T^{-3}]$, and can be expressed as the dot product of force and velocity vectors ($P = \vec{F} \cdot \vec{V}$).

12. **Answer :** A, B, D

Statements A, B, and D provide correct formulas and unit conversions for power.

Statement C is incorrect for power.

ASSERTION AND REASON TYPE

13. **Answer :** A) Both A and R are true and R is the correct explanation of A

Both statements are factually correct.

Since gravity pulls the object downward, energy must be supplied to lift it against this force. This energy is stored as gravitational potential energy. Therefore, the reason correctly explains the assertion.

14. **Answer :** C) A is false but R is true

The assertion is false because the power of a machine depends on both the work done and the time taken to do it.

The reason provides the correct definition of power.

COMPREHENSION TYPE

15. **Answer :** A) 2.55 m

$m = 5.8$ kg

$P.E = 145$ J

Assuming $g = 9.8$ m/s^2:

$P.E = m \cdot g \cdot h$

$145 = 5.8 \cdot 9.8 \cdot h$

$h = \frac{145}{56.84}$

$h = 2.55$ m

16. **Answer :** B) 4500 J

$m = 30$ kg

$h = 15$ m

$g = 10$ m/s^2

$P.E = m \cdot g \cdot h$

$P.E = 30 \cdot 10 \cdot 15$

$P.E = 4500$ J

17. **Answer :** B) 6000 J

$m = 50$ kg

$h = 12$ m

Assuming $g = 10$ m/s^2:

$P.E = m \cdot g \cdot h$

$P.E = 50 \cdot 10 \cdot 12$

$P.E = 6000$ J

18. **Answer :** A) 600 W

$m = 90$ kg

$h = 10$ m

$t = 15$ s

Assuming $g = 10$ m/s^2:

$W = m \cdot g \cdot h$

$W = 90 \cdot 10 \cdot 10$

$W = 9000$ J

$P_1 = \frac{9000}{15}$

$P_1 = 600$ W

19. **Answer :** D) 450 W

$W = 9000$ J

$t = 20$ s

$P_2 = \frac{9000}{20}$

$P_2 = 450$ W

20. **Answer :** C) $\frac{4}{3}$

$Ratio = \frac{P_1}{P_2}$

$Ratio = \frac{600}{450}$

$Ratio = \frac{4}{3}$

MATRIX MATCHING TYPE

21. **Answer :** a → Q, b → Q or R, c → P, d → R

Force is the negative gradient of potential energy ($F = -\frac{dU}{dx}$).

At point A, the slope is positive, so Force is negative (Q).

At point B (before peak), the slope is positive, so Force is negative (Q).

At point B (peak), the slope is zero, so Force is zero (R).

At point C, the slope is negative, so Force is positive (P).
At point E (valley), the slope is zero, so Force is zero (R).

22. **Answer :** a → S, b → R, c → Q, d → P
Power is a scalar quantity (S).
The dimensional formula of power is $[ML^2T^{-3}]$ (R).
1 HP is equal to 746 W (Q).
The C.G.S unit of power is erg/sec (P).

INTEGER TYPE

23. **Answer :** 375 W or 367.5 W
$m = 50$ kg
Total height $h = 45 \text{ steps} \cdot 0.15 \text{ m/step} = 6.75$ m
$t = 9$ s
Assuming $g = 10 \text{ m/s}^2$:
$W = m \cdot g \cdot h$
$W = 50 \cdot 10 \cdot 6.75$
$W = 3375$ J
$P = \frac{W}{t}$
$P = \frac{3375}{9}$
$P = 375$ W
(Using $g = 9.8 \text{ m/s}^2$ gives exactly 367.5 W*)*

24. **Answer :** 234 kg
$P = 0.39$ kW $= 390$ W
$h = 10$ m
$t = 60$ s
Assuming $g = 10 \text{ m/s}^2$:
$P = \frac{m \cdot g \cdot h}{t}$
$390 = \frac{m \cdot 10 \cdot 10}{60}$
$390 = \frac{m \cdot 100}{60}$
$m = \frac{390 \cdot 60}{100}$
$m = 234$ kg

25. **Answer :** 200 W or 196 W
Mass lifted per second $\frac{m}{t} = 2$ kg/s
$h = 10$ m
Assuming $g = 10 \text{ m/s}^2$:
$P = \frac{m \cdot g \cdot h}{t}$
$P = \left(\frac{m}{t}\right) \cdot g \cdot h$
$P = 2 \cdot 10 \cdot 10$
$P = 200$ W
(Using $g = 9.8 \text{ m/s}^2$ gives exactly 196 W*)*

5. Heat

Heat, Temperature and Thermometers

LEVEL-1

1. **Answer :** A) Hot, cold
Heat is always transmitted from a hot substance to a cold substance.

2. **Answer :** C) It does not contain mercury
Digital thermometers are safer because they do not contain mercury, which is a toxic liquid metal.

3. **Answer :** B) Scalar
Temperature only has magnitude and no direction, making it a scalar quantity.

4. **Answer :** A) Laboratory thermometer
A laboratory thermometer is generally used to measure the temperature of different things and places in science experiments.

5. **Answer :** C) Bulb
The bulb of the thermometer contains the mercury and must be in direct contact with the object to measure its temperature accurately.

6. **Answer :** C) 212
The boiling point of water is 100°C, which corresponds to 212°F on the Fahrenheit scale.

7. **Answer :** B) 180
The Lower Fixed Point (LFP) is 32°F and the Upper Fixed Point (UFP) is 212°F.
Divisions = $212 - 32$
Divisions = 180

8. **Answer :** B) 12.5°C
The relation between the Reaumur and Celsius scales is:
$\frac{C}{100} = \frac{R}{80}$
$\frac{C}{5} = \frac{R}{4}$
$C = \frac{5}{4} \cdot 10$
$C = 12.5°C$

9. **Answer :** C) 312.4 K
First, convert Fahrenheit to Celsius:
$C = \frac{5}{9} \cdot (F - 32)$
$C = \frac{5}{9} \cdot (103 - 32)$
$C = \frac{5}{9} \cdot 71$
$C \approx 39.44°C$
Now, convert Celsius to Kelvin:
$K = C + 273$
$K = 39.44 + 273$
$K = 312.44$ K

10. **Answer :** C) 131°F

$F = \frac{9}{5} \cdot C + 32$

$F = \frac{9}{5} \cdot 55 + 32$

$F = 99 + 32$

$F = 131°F$

11. **Answer :** C) 3 kcal

We know that 1 calorie = 4.2 J.

Heat in calories = $\frac{12600}{4.2}$

Heat in calories = 3000 cal

Heat in kilocalories = 3 kcal

12. **Answer :** B) 35°C

$C = \frac{5}{9} \cdot (F - 32)$

$C = \frac{5}{9} \cdot (95 - 32)$

$C = \frac{5}{9} \cdot 63$

$C = 35°C$

13. **Answer :** C) 100°C

212°F is the boiling point of water, which exactly corresponds to 100°C on the Celsius scale.

14. **Answer :** B) 60°C

Using the formula for faulty thermometers:

$\frac{\text{Observed} - \text{Marked LFP}}{\text{Marked UFP} - \text{Marked LFP}} = \frac{C - 0}{100}$

$\frac{70 - 10}{110 - 10} = \frac{C}{100}$

$\frac{60}{100} = \frac{C}{100}$

$C = 60°C$

15. **Answer :** D) Both readings represent the same temperature.

Convert 313 K to Celsius:

$C = 313 - 273$

$C = 40°C$

Convert 40°C to Fahrenheit:

$F = \frac{9}{5} \cdot 40 + 32$

$F = 72 + 32$

$F = 104°F$

Both values are equivalent.

16. **Answer :** D) 81°F

Calculate the change in Celsius temperature:

$\Delta C = 80 - 35$

$\Delta C = 45°C$

Convert the temperature difference to Fahrenheit:

$\Delta F = \frac{9}{5} \cdot \Delta C$

$\Delta F = \frac{9}{5} \cdot 45$

$\Delta F = 81°\text{F}$

17. **Answer :** C) $50°\text{C}, 323\ \text{K}$

Using the formula for faulty thermometers:

$\frac{\text{Observed} - \text{Marked LFP}}{\text{Marked UFP} - \text{Marked LFP}} = \frac{C - 0}{100}$

$\frac{70 - 20}{120 - 20} = \frac{C}{100}$

$\frac{50}{100} = \frac{C}{100}$

$C = 50°\text{C}$

Convert to Kelvin:

$K = 50 + 273$

$K = 323\ \text{K}$

18. **Answer :** C) 40°C

$C = \frac{5}{9} \cdot (F - 32)$

$C = \frac{5}{9} \cdot (104 - 32)$

$C = \frac{5}{9} \cdot 72$

$C = 40°\text{C}$

LEVEL-II

SINGLE CORRECT OPTION TYPE

1. **Answer :** B) 50°C

Using the formula for faulty thermometers:

$\frac{\text{Observed} - \text{Marked LFP}}{\text{Marked UFP} - \text{Marked LFP}} = \frac{C - 0}{100}$

$\frac{83 - 8}{158 - 8} = \frac{C}{100}$

$\frac{75}{150} = \frac{C}{100}$

$0.5 = \frac{C}{100}$

$C = 50°\text{C}$

2. **Answer :** C) 81°F

The change in Fahrenheit is given by:

$\Delta F = \frac{9}{5} \cdot \Delta C$

$\Delta F = \frac{9}{5} \cdot 45$

$\Delta F = 81°\text{F}$

3. **Answer :** B) 40°C

Using the formula for faulty thermometers:

$\frac{68 - 20}{140 - 20} = \frac{C}{100}$

$\frac{48}{120} = \frac{C}{100}$

$0.4 = \frac{C}{100}$

$C = 40°C$

4. **Answer :** C) 55°

First, find the true final equilibrium temperature. Mixing equal masses of water at 20°C and 80°C gives:

$T_{true} = \frac{20 + 80}{2} = 50°C$

Now, find the reading on the faulty thermometer:

$\frac{\text{Observed} - 5}{105 - 5} = \frac{50}{100}$

$\frac{\text{Observed} - 5}{100} = 0.5$

$\text{Observed} - 5 = 50$

$\text{Observed} = 55°$

5. **Answer :** C) 333 K

The range between ice point and steam point is 100°C.

150 divisions = 100°C

1 division $= \frac{100}{150} = \frac{2^{\circ}}{3}$ C

90 divisions $= 90 \cdot \frac{2}{3} = 60°C$

Convert to Kelvin:

$K = 60 + 273 = 333$ K

6. **Answer :** C) 30 K

A change in temperature on the Kelvin scale equals the change on the Celsius scale.

$\Delta K = \Delta C = \frac{5}{9} \cdot \Delta F$

$\Delta K = \frac{5}{9} \cdot 54$

$\Delta K = 30$ K

7. **Answer :** invalid question (reading should be 56.4° instead of 102°)

First, find the actual temperature in Celsius using the faulty thermometer formula:

$\frac{56.4 - 12}{132 - 12} = \frac{C}{100}$

$0.37 = \frac{C}{100}$

$C = 37°C$

Now, convert this true temperature to Fahrenheit:

$F = \frac{9}{5} \cdot 37 + 32$

$F = 66.6 + 32$

$F = 98.6°F$

8. **Answer :** B) 131°F

First, find the equilibrium temperature of the mixture:

$T_{mix} = \frac{40 + 100}{2} = 70°C$

The mixture is cooled by 15°C:

$T_{final} = 70 - 15 = 55°C$

Convert to Fahrenheit:

$F = \frac{9}{5} \cdot 55 + 32$

$F = 99 + 32 = 131°F$

9. **Answer :** B) 333 K

Using the formula for faulty thermometers:

$\frac{80 - (-10)}{140 - (-10)} = \frac{C}{100}$

$\frac{90}{150} = \frac{C}{100}$

$0.6 = \frac{C}{100}$

$C = 60°C$

Convert to Kelvin:

$K = 60 + 273 = 333$ K

ONE OR MORE THAN ONE CORRECT OPTION TYPE

10. **Answer :** A, B, D

Calculate the true temperature in Celsius:

$\frac{85 - 10}{160 - 10} = \frac{C}{100}$

$\frac{75}{150} = \frac{C}{100}$

$C = 50°C$

Evaluating the statements:

A) True temperature is 50°C (Correct)

B) True temperature in Kelvin is $50 + 273 = 323$ K (Correct)

C) The reading 85° is exactly at the midpoint of the scale, but the *reading value* (85) is not mathematically half of the scale's *range* (150). (False)

D) True Fahrenheit temperature is $\frac{9}{5} \cdot 50 + 32 = 122°F$ (Correct)

11. **Answer :** A, B, D

Convert 327 K to Celsius:

$C = 327 - 273 = 54°C$

Convert to Fahrenheit:

$F = \frac{9}{5} \cdot 54 + 32 = 129.2°F$

Evaluating the statements:

A) Temperature in Celsius is 54°C (Correct)

B) Temperature in Fahrenheit is 129.2°F (Correct)

C) 54°C is below the boiling point of water (100°C) (False)

D) 54°C is above normal human body temperature (37°C) (Correct)

12. **Answer :** A, B, C

Convert Joules to calories (1 cal = 4.2 J):

Heat (cal) $= \frac{25200}{4.2} = 6000$ cal

Convert to kilocalories:

Heat (kcal) $= \frac{6000}{1000} = 6$ kcal

Convert Joules to kilojoules:

Heat (kJ) $= \frac{25200}{1000} = 25.2$ kJ

Statements A, B, and C are all mathematically correct.

13. **Answer :** A, B, C, D
Calculate the fall in Celsius:
$\Delta C = 80 - 20 = 60°C$
(A is correct)
Calculate the fall in Kelvin (which is equal to the fall in Celsius):
$\Delta K = 60$ K
(B is correct)
Calculate the fall in Fahrenheit:
$\Delta F = \frac{9}{5} \cdot 60 = 108°F$
(C is correct)
Calculate the final temperature in Kelvin:
$K_{final} = 20 + 273 = 293$ K
(D is correct)

ASSERTION AND REASON TYPE

14. **Answer :** A) Both A and R are true, and R is the correct explanation of A.
Heat naturally flows from a body at a higher temperature to a body at a lower temperature. Temperature acts as the thermal potential that dictates this direction.

15. **Answer :** A) Both A and R are true, and R is the correct explanation of A.
Alcohol freezes at a much lower temperature than mercury (around $-115°C$ vs $-39°C$), which strictly makes alcohol thermometers suitable for measuring very low temperatures.

MATRIX MATCHING TYPE

16. **Answer :** A → R, B → P, C → Q, D → S
Celsius Scale: LFP = 0°, UFP = 100°, 100 divisions (R).
Fahrenheit Scale: LFP = 32°, UFP = 212°, 180 divisions (P).
Kelvin Scale: LFP = 273, UFP = 373, 100 divisions (Q).
Reaumur Scale: LFP = 0°, UFP = 80°, 80 divisions (S).

17. **Answer :** A → Q, B → S, C → P, D → R
Clinical Thermometer: Used for measuring body temperature and contains a kink to hold the reading (Q).
Laboratory Thermometer: Used for measuring the temperature of liquids and chemicals (S).
Six's Maximum-Minimum Thermometer: Used for recording the highest and lowest temperature of a day (P).
Digital Thermometer: Safer because it does not contain mercury (R).

COMPREHENSION TYPE

18. **Answer :** B) Laboratory thermometers do not have a kink to hold the reading.
Because laboratory thermometers lack a kink, the liquid level drops immediately when removed from the hot object, so they must be read while immersed.

19. **Answer :** C) Constriction or kink
The constriction (kink) prevents the mercury column from falling back immediately, allowing the user to note the reading conveniently after removing it.

20. **Answer :** C) 39°C
$C = \frac{5}{9} \cdot (102.2 - 32)$
$C = \frac{5}{9} \cdot 70.2$
$C = 39°C$

21. **Answer :** C) 60°C
Mixing equal masses of the same substance at different temperatures results in the average temperature:
$T_{final} = \frac{90 + 30}{2}$
$T_{final} = \frac{120}{2}$
$T_{final} = 60°C$

22. **Answer :** C) 333 K
$K = C + 273$
$K = 60 + 273$
$K = 333$ K

23. **Answer :** D) 100°
Using the formula for faulty thermometers:
$\frac{\text{Observed} - 10}{160 - 10} = \frac{60}{100}$
$\frac{\text{Observed} - 10}{150} = 0.6$
$\text{Observed} - 10 = 0.6 \cdot 150$
$\text{Observed} - 10 = 90$
$\text{Observed} = 100°$

INTEGER TYPE

24. **Answer :** 20
$C = \frac{5}{9} \cdot (F - 32)$
$C = \frac{5}{9} \cdot (68 - 32)$
$C = \frac{5}{9} \cdot 36$
$C = 20°C$

25. **Answer :** 40
$C = K - 273$
$C = 313 - 273$
$C = 40°C$

26. **Answer :** 50
Mixing equal quantities of water at different temperatures yields the average:
$T_{final} = \frac{80 + 20}{2}$
$T_{final} = 50°C$

Transmission of Heat, Effects of Heat and Water Cycle

LEVEL-1

1. **Answer :** B) Heat flows from spoon to water by conduction
Heat spontaneously transfers from a hotter object (the spoon) to a colder object (the water). Because the spoon is a solid metal, this transfer of thermal energy occurs via conduction.

2. **Answer :** C) Heat transfer takes place due to movement of particles
The upward movement of coloured, heated water and downward movement of cooler water visually demonstrates convection currents, where heat is transferred due to the actual macroscopic movement of the liquid particles themselves.

3. **Answer :** C) Feeling warm under sunlight

The thermal energy from the sun reaches the earth through the vacuum of space. Heat transfer by radiation does not require any material medium.

4. **Answer :** C) Land heats faster and air above it rises
During daytime, land heats up much faster than the sea. The warm air above the land rises (becoming less dense), and the cooler, denser air from the sea rushes in to take its place, creating a sea breeze.

5. **Answer :** C) Is a good conductor and transfers heat quickly
Copper is an excellent thermal conductor, allowing it to efficiently transfer excess heat away from the engine to the surrounding air.

6. **Answer :** C) Black absorbs more thermal radiation
Black or dark-coloured surfaces are excellent absorbers of thermal radiation compared to white or shiny surfaces, causing the water inside to heat up faster.

7. **Answer :** C) It requires movement of medium particles
Convection is the mode of heat transfer where the actual movement of the medium's particles (in fluids like liquids and gases) carries the thermal energy.

8. **Answer :** C) Air above sea becomes warmer and rises
At night, the land cools down much faster than the sea. Therefore, the air above the relatively warmer sea rises, and the cooler, denser air from the land flows out to the sea, creating a land breeze.

9. **Answer :** B) Prevent conduction and convection
A vacuum completely lacks material particles. Since both conduction and convection require a material medium to transfer heat, the vacuum layer effectively prevents heat loss by these two modes.

10. **Answer :** C) Prevent heat transfer by radiation
Shiny, silvered surfaces are excellent reflectors and very poor emitters of thermal radiation. They reflect the radiant heat back into the flask, minimizing heat loss by radiation.

11. **Answer :** B) Mercury expands on heating
Liquids expand when heated. The hot tea transfers heat to the thermometer bulb, causing the mercury inside to expand and rise up the capillary tube.

12. **Answer :** C) Changing the state of ice into water
During a phase change (like melting), the supplied thermal energy is entirely utilized to overcome intermolecular forces and increase the distance between molecules, changing the state from solid to liquid without increasing the temperature.

13. **Answer :** B) Evaporation
Evaporation is a slow, gradual surface phenomenon that can occur at any temperature below the boiling point. Boiling is a rapid, bulk process.

14. **Answer :** C) Sublimation
Sublimation is the direct phase transition from a solid to a gas without passing through the intermediate liquid phase.

15. **Answer :** C) Linear expansion
The expansion of a solid object predominantly in one dimension (length) when heated is termed linear expansion.

16. **Answer :** C) Transpiration
Transpiration is the biological process by which moisture is carried through plants from roots to small pores on the underside of leaves, where it changes to vapour and is released into the atmosphere.

17. **Answer :** C) Infiltration
Infiltration is the physical process by which water on the ground surface enters and moves downward through the soil and porous rocks.

18. **Answer :** C) Gravel
Gravel has the largest particle size and the largest continuous void spaces between particles, offering the least resistance to the downward flow of water compared to sand, clay, or solid rock.

LEVEL-II

SINGLE CORRECT OPTION TYPE

1. **Answer :** C) Conduction
In solids like a copper rod, heat is primarily transferred from the hotter end to the colder end through molecular vibrations and free electrons without the actual movement of the material itself. This is conduction.

2. **Answer :** C) Conducts heat quickly
Copper is an excellent thermal conductor. It is used at the bottom of cooking vessels to distribute heat rapidly and evenly across the base.

3. **Answer :** C) More than shiny surface
A black surface is an ideal absorber of radiant heat, whereas a shiny surface is a good reflector and a poor absorber.

4. **Answer :** C) Vacuum prevents conduction and convection
Conduction and convection both require a material medium. The vacuum layer between the walls of a thermos flask lacks particles, thus effectively blocking heat transfer via these two modes.

5. **Answer :** B) 4%
If length L and breadth B both increase by 2% (i.e., new dimensions are $1.02L$ and $1.02B$).
New Area = $1.02L \cdot 1.02B$
New Area = $1.0404 \cdot (L \cdot B)$
This represents an approximate area increase of 4%.

6. **Answer :** C) Convection
Convection relies strictly on the macroscopic movement of fluid particles (liquids or gases). Because a vacuum has no particles, convection cannot occur.

7. **Answer :** C) Reflects most radiation
White or light-coloured surfaces are good reflectors and poor absorbers of thermal radiation. LPG tanks are painted white to prevent them from absorbing excessive heat from the sun.

8. **Answer :** B) Air above sea becomes warmer and rises
During the night, the land cools down faster than the sea. The relatively warmer air above the sea expands and rises, while cooler air from the land rushes in to fill the void, creating a land breeze.

9. **Answer :** B) Deposition
Deposition is the thermodynamic process where a gas transforms directly into a solid state without passing through the liquid phase.

10. **Answer :** C) Increase the distance between molecules
During a change of state at a constant temperature (like melting or boiling), the supplied latent heat is completely used to overcome the cohesive forces holding the molecules together, thereby increasing intermolecular distance.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Answer :** A, B, C, D
Thermal radiation is an electromagnetic wave. Therefore, it can travel through a vacuum at the speed of light. Dark surfaces absorb more radiation, while white/shiny surfaces reflect it.

12. **Answer :** A, B, C
Melting (solid to liquid), boiling (liquid to gas), and sublimation (solid to gas) are endothermic processes that require the absorption of heat. Condensation releases heat.

13. **Answer :** A, B, C
Infiltration depends on the porosity and permeability of the material. This is dictated by particle size, how tightly they are packed, and the resulting void spaces. The colour of rocks has no physical effect on water flow.

14. **Answer :** A, B, C, D
Convection requires the bulk movement of the heated fluid matter. This occurs freely in liquids and gases but cannot occur in rigid solids. The process naturally establishes circulating convection currents.

ASSERTION AND REASON TYPE

15. **Answer :** A) Both A and R are true and R is correct explanation
A perfect black body is both an ideal absorber and an ideal emitter (radiator) of thermal radiation. Therefore, a black surface will absorb heat quickly when cold, but will also radiate heat away quickly to cool down when hot.

16. **Answer :** A) Both A and R are true and R is correct explanation
During boiling, the temperature remains constant at the boiling point. The thermal energy supplied (latent heat of vaporization) is utilized entirely to break intermolecular bonds and increase the separation between molecules to change the liquid into a gas.

MATRIX MATCHING TYPE

17. **Answer :** A → Q, B → P, R, C → P, D → Q, S
Melting: Requires the absorption of heat (Q).
Condensation: Is the phase change from Gas to Liquid (R) and releases heat (P).
Freezing: Releases heat as the liquid solidifies (P).
Sublimation: Is the phase change from Solid directly to Gas (S) and requires the absorption of heat (Q).

18. **Answer :** A → Q, B → R, C → P, D → S
Heating an iron rod: Heat transfers through the solid metal by Conduction (Q).
Sea breeze: This is a natural example of fluid movement driven by heat, which is Convection (R).
Sun heating Earth: Heat travels through the vacuum of space via Radiation (P).
Thermos silvering: The shiny walls prevent heat loss through the Reflection of thermal radiation (S).

COMPREHENSION TYPE

19. **Answer :** B) Sea breeze
The daytime flow of cooler air from the sea toward the relatively hotter land is known as a sea breeze.

20. **Answer :** C) Density decreases
When the air above the land heats up, it expands. Its volume increases while its mass remains constant, causing its overall density to decrease, which makes it rise.

21. **Answer :** C) Convection
The formation of land and sea breezes is a classic natural example of convection currents established in the atmosphere due to differential heating.

22. **Answer :** C) Radiation
Since the vessels are identical except for their colour/surface finish, and both contain hot water, the primary mode of heat loss that differentiates them is thermal radiation.

23. **Answer :** C) Black vessel
Black surfaces are excellent absorbers of thermal radiation. When placed in sunlight, the black vessel will absorb heat radiation much faster than the white, reflective vessel.

24. **Answer :** C) Black surfaces are good absorbers and radiators
Based on the observations, the black vessel cooled faster (good radiator) and will heat up faster (good absorber). Good absorbers of radiation are simultaneously good radiators.

INTEGER TYPE

25. **Answer :** 10
Original length = 250 cm
Increase = 4% of 250
$\text{Increase} = \frac{4}{100} \cdot 250$
Increase = 10 cm

26. **Answer :** 85
The question asks only for the change in temperature during the *heating* phase.
Change = 100°C − 15°C
Change = 85°C

27. **Answer :** 4
$\text{Ratio} = \frac{\text{Heat absorbed by black}}{\text{Heat absorbed by white}}$
$\text{Ratio} = \frac{96}{24}$
Ratio = 4

Previous Year Questions

1. **Answer :** C) Rod Y being a good conductor of heat, conducts away the heat given to the paper
Rod Y is made of a material that is a good thermal conductor (like a metal). It quickly conducts the heat away from the paper, preventing the paper from reaching its ignition temperature. Rod X is a bad conductor, so heat accumulates at the paper until it burns.

2. **Answer :** D) II, III and IV only
I is incorrect: A thermometer should never be held by its bulb while reading, as body heat will alter the reading.
II is correct: Sharp jerks are given to a clinical thermometer to force the mercury level down past the kink before a new measurement.
III is correct: Black fabric absorbs more radiant heat from the sun, keeping the body warmer in winter.
IV is correct: Jute sacks act as thermal insulators, trapping air and preventing ambient heat from rapidly melting the ice.

3. **Answer :** A) (i) and (ii) only

(i) is correct: Because the paper is thin, it offers little thermal resistance, allowing heat to rapidly conduct through it into the water.

(ii) is correct: Water boils at 100°C, which is significantly lower than the ignition temperature of paper (around 233°C). As long as there is water to absorb the heat and boil, the paper's temperature cannot rise high enough to catch fire.

(iii) is incorrect: If the paper were thick, it would act as an insulator, trap heat, and eventually burn.

4. **Answer :** C) $1.0°C$ lower than the true temperature

Let T_{true} be the true temperature and T_{faulty} be the reading. Using the faulty thermometer formula:

$$\frac{T_{faulty} - \text{LFP}_{faulty}}{\text{UFP}_{faulty} - \text{LFP}_{faulty}} = \frac{T_{true} - \text{LFP}_{true}}{\text{UFP}_{true} - \text{LFP}_{true}}$$

$$\frac{T_{faulty} - (-1)}{99 - (-1)} = \frac{T_{true} - 0}{100}$$

$$\frac{T_{faulty} + 1}{100} = \frac{T_{true}}{100}$$

$$T_{faulty} + 1 = T_{true}$$

$$T_{faulty} = T_{true} - 1.0$$

The measured temperature is 1.0°C lower than the true temperature.

5. **Answer :** B) (i) and (iv) only

(ii) is incorrect: Matter universally expands on heating, it does not contract.

(iii) is incorrect: Both identical steel bars P and Q will expand and increase in length when heated.

(i) is correct: Because both bars expand towards each other, the 1 cm gap between them will decrease.

(iv) is correct: Metals are excellent conductors of heat.

6. **Answer :** A) (ii) and (iv) only

Convection in fluids occurs when the fluid is heated from *below*. The heated fluid expands, becomes less dense, and rises, while cooler, denser fluid sinks.

In **(ii)**, the electric heater is at the bottom. Water is heated from below, establishing proper convection currents.

In **(iv)**, the candle flame heats the beaker from the bottom, establishing proper convection currents.

In **(i)** and **(iii)**, the heat source is at the top. The water at the top heats up, becomes less dense, and simply stays at the top. Convection currents cannot form downwards.

6. Electricity And Circuits

LEVEL-1

1. **Answer :** C) Electric cell
An electric cell is a device that converts stored chemical energy into electrical energy to provide electric current.

2. **Answer :** C) Filament
The spirally wound wire inside an electric bulb that glows when heated is called the filament.

3. **Answer :** D) Tungsten
The filament in a bulb is typically made of tungsten because it has a very high melting point and can glow brightly without melting.

4. **Answer :** A)
The standard electrical symbol for a bulb is a circle with a filament symbol inside it.

5. **Answer :** B) 2
An electric cell has two terminals: a positive terminal and a negative terminal.

6. **Answer :** C) Switch (or) key
A switch or key is the component used to either make (close) or break (open) an electric circuit.

7. **Answer :** C) Metal cap
In a standard electric cell, the metal cap acts as the positive terminal.

8. **Answer :** B) Closed
A complete and continuous path for electricity to flow is called a closed electric circuit.

9. **Answer :** C) Positive to negative
By convention, electric current in a closed circuit always flows from the positive terminal of the cell to the negative terminal.

10. **Answer :** C) Electric current
Electric current is the form of electrical energy that continuously flows through a closed circuit.

11. **Answer :** D) Resistance
Resistance is the inherent property of a material to oppose or obstruct the flow of electrons.

12. **Answer :** C) Nichrome
Nichrome is an alloy specifically known for having a very high electrical resistance.

13. **Answer :** C)
The standard circuit symbol for an electrical resistor is a zigzag line.

14. **Answer :** D) Resistance
The friction or obstruction offered to the passage of electric current by any material is defined as its resistance.

15. **Answer :** B) Conductors
Substances that have a large number of free electrons allow current to flow easily and are called conductors.

16. **Answer :** D) Silver & Rubber
Silver is an excellent conductor of electricity, while rubber is an excellent insulator.

17. **Answer :** B) Insulators
Materials which do not allow electricity to pass through them easily are called insulators.

18. **Answer :** D) Rubber
Rubber does not allow electric current to pass through it, making it an insulator.

19. **Answer :** D) Plastic
Electrical wires are covered with plastic because it is an insulator, which protects us from electric shocks.

20. **Answer :** D) Metals are conductors
Metals are used to make electrical wires because they are good conductors and allow electricity to flow easily.

21. **Answer :** B) Series circuit
An electric circuit in which all components are connected one after the other in a single continuous path is called a series circuit.

22. **Answer :** C) Electric current
In a series connection, the electric current has only one path to flow, so it remains the same through all components.

23. **Answer :** A) Voltage
In a parallel connection, all branches are connected across the same two points, so the voltage remains constant across each branch.

24. **Answer :** D) Battery
A battery is formed by the combination of two or more electric cells connected together.

25. **Answer :** B) The thin filament
Light is produced in an electric bulb when the electric current heats the thin filament until it glows.

LEVEL-II

SINGLE CORRECT OPTION TYPE

1. **Answer :** D) All bulbs will go together
When the switch is moved to the ON position, the circuit is completed and current flows through the entire circuit almost instantly, causing all bulbs to glow together.

2. **Answer :** D) A thin wire
The filament of a torch bulb is a very thin wire made of tungsten that glows when heated by an electric current.

3. **Answer :** D) Steel spoon
To complete an electric circuit, a conductive material is needed.
A steel spoon is made of metal and is a good conductor of electricity.

4. **Answer :** B) Bad conductors
Materials that offer high resistance strictly oppose the flow of electric current and are therefore called bad conductors.

5. **Answer :** D) Circuit
A closed and continuous path through which electric current flows is called an electric circuit.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

6. **Answer :** A, C
A bulb can be made to glow by completing the circuit with any conductive material.
Both copper and an iron nail are good conductors of electricity.

7. **Answer :** B, C
The filament in a bulb must withstand very high temperatures to glow without melting, so it has a high melting point.
It also has higher electrical resistance so that it generates enough heat when current passes through it.

8. **Answer :** B, C
A switch is an essential component used to either make (close) or break (open) an electric circuit.

9. **Answer :** B, C
In a parallel circuit connection, the total current from the source is divided among the parallel branches.
However, the voltage across each parallel branch remains constant and equal to the source voltage.

COMPREHENSION TYPE

10. **Answer :** B) 2
An electric cell has exactly two terminals: a positive terminal and a negative terminal.

11. **Answer :** D) Chemical to electric
An electric cell contains chemicals and converts this stored chemical energy into electrical energy.

12. **Answer :** D) Metal disc
In a standard electric cell, the flat metal disc at the bottom acts as the negative terminal.

13. **Answer :** C) Electric current
The net flow of electric charge (electrons) through a conducting wire in a given time interval is defined as the electric current.

14. **Answer :** A) Resistance
The opposition or obstruction encountered by electrons as they flow through a wire is called resistance.

15. **Answer :** D) Ohmmeter
An ohmmeter is the specific measuring instrument used to measure the electrical resistance of a component.

ASSERTION AND REASON TYPE

16. **Answer :** Both A and R are false (No correct option provided)
An ammeter is used to measure the electric current flowing through a circuit.
A voltmeter is used to measure the potential difference (voltage) in a circuit.
Therefore, both the Assertion and the Reason are false.

17. **Answer :** A) Both A and R are true, and R is the correct explanation of A
Earthing provides safety by directing any excess or leakage electric charge safely into the ground through a low-resistance conductor, preventing electric shocks.

MATRIX MATCHING TYPE

18. **Answer :** a → S, b → T, c → R, d → Q
a) → S) Open circuit
b) → T) Variable resistor
c) → R) Bulb
d) → Q) Closed circuit

19. **Answer :** a → Q, b → R, c → P, d → S, T
In a parallel circuit, the voltage across each branch remains constant (Q).
The metal cap of an electric bulb acts as the positive terminal (R).

The metal disc at the base of the bulb acts as the negative terminal (P).
In a series circuit, the current flowing through all components remains constant (S) and voltage is divided (T).

INTEGER TYPE

20. **Answer :** 3
In a series connection, the voltage of the cell is divided equally among all the identical bulbs.
Since 3 bulbs are connected in series to the same cell, each bulb receives only one-third of the total voltage.
Therefore, all 3 bulbs will glow dimly compared to a single bulb connected to the cell.

21. **Answer :** 3
The initial number of resistors is 2.
The final number of resistors is 5.
Increase = $5 - 2$
Increase = 3

22. **Answer :** 3
In a parallel circuit, each component operates independently of the others.
If 1 bulb is removed, the remaining bulbs continue to receive the full voltage from the source.
Remaining glowing bulbs = $4 - 1$
Remaining glowing bulbs = 3

PREVIOUS YEAR QUESTIONS

1. **Answer :** B) III only
A plastic disposable spoon is an insulator.
Placing it in the series circuit prevents the flow of electric current, meaning none of the bulbs will light up.

2. **Answer :** (i) F, (ii) T, (iii) T
Circuit I: Four batteries are connected in parallel, so the effective voltage is E → brightness $\propto E^2/R$.
Circuit II: Four batteries are connected in series, so the effective voltage is 4E → brightness $\propto (4E)^2/R = 16E^2/R$(brightest).
Circuit III: Two branches of 2 batteries in series are connected in parallel, giving an effective voltage of 2E → brightness $\propto 4E^2/R$.
Circuit IV: One battery on the left and three at the bottom are all in series, giving 4E, the same as Circuit II → brightness $\propto 16E^2/R$.
Therefore:
(i) False
(ii) True
(iii) False

3. **Answer :** D) X, Y and Z
Closing the switch completes the single continuous loop of the circuit.
Since all three bulbs X, Y, and Z are connected in this closed series loop, all of them will receive current and glow.

4. **Answer :** D) Dimmer Dimmer Brighter Dimmer
Brightness of the Bulbs
Bulb B_1: It carries the total circuit current. As the total current decreases, the power dissipated by B_1decreases. Hence, B_1becomes dimmer.

Bulbs $B_3$and B_5: These bulbs remain connected in parallel and continue to share the current equally. Since the total current entering this branch is now smaller, the current through each bulb decreases. Therefore, both $B_3$and B_5become dimmer.
Bulb B_4: Initially, the current in the left branch was shared equally between $B_2$and B_4. After B_2blows, the entire branch current flows only through B_4. Although the overall circuit current decreases slightly, B_4no longer shares the current with another bulb. As a result, the current through B_4increases, so its power increases and B_4becomes brighter.

5. **Answer :** C) Walkman Table lamp Electric kettle
Based on the classification, X represents devices producing sound, Y represents devices producing light, and Z represents devices producing heat.
A walkman produces sound, a table lamp produces light, and an electric kettle produces heat.

6. **Answer :** B) If bulb Z is removed, then bulb X will glow brighter.
Bulbs X, Y, and Z are all connected in a single series loop with the battery.
If bulb Z is removed, the entire circuit is broken and no current will flow.
Therefore, bulb X will not glow at all, making statement B incorrect.

7. **Answer :** C) Both (i) and (ii) are true
An electric cell generates electricity by converting stored chemical energy into electrical energy.
A battery is defined as a combination of two or more electric cells connected together.

8. **Answer :** B) Only B_1 and B_3 will glow
Bulb B_1 is in the main circuit line, while B_2 and B_3 are in parallel branches.
When switch S_2 is closed and S_1 is open, current flows through the bottom branch but not the middle branch.
Therefore, only bulbs B_1 and B_3 will receive current and glow.

9. **Answer :** C) 2 and 3 only
Bulbs 2 and 3 are located in the rightmost parallel branch, which is controlled by switch S.
Bulb 1 is in a separate parallel branch connected before the switch.
When switch S is opened, only the right branch is disconnected, so bulbs 2 and 3 will switch off.

10. **Answer :** B) 2, 4, 5 and 6 only
Statement 2 describes the switch, statement 4 describes the contact springs, statement 5 describes the outer protective casing, and statement 6 describes the bulb.
Statement 1 is incorrect as reflectors spread light over a larger distance, and statement 3 is incorrect as an electric cell converts chemical energy into electrical energy, not the other way around.

7. Magnetism

LEVEL-I

1. **Answer :** D) All of these
Bar magnets, horseshoe magnets, and magnetic needles are all examples of artificial magnets made by humans.

2. **Answer :** A) Aluminium, nickel and cobalt
The name "Alnico" itself is an acronym derived from its primary elemental components: Aluminium (Al), Nickel (Ni), and Cobalt (Co).

3. **Answer :** B) Fe_3O_4
The chemical formula for magnetite, which is a naturally occurring magnet (lodestone), is Fe_3O_4.

4. **Answer :** A) Para magnetic
Materials that are weakly attracted by magnets are classified as paramagnetic materials.

5. **Answer :** C) Both A & B
Both manganese and platinum are weakly attracted to external magnetic fields, making them paramagnetic materials.

6. **Answer :** A) Iron
Iron is strongly attracted by a magnet and can be easily magnetized, classifying it as a ferromagnetic material.

7. **Answer :** C) Less than
The magnetic poles reside slightly inside the ends of a magnet.
Therefore, the magnetic length is always slightly less than the overall geometric length of the magnet.

8. **Answer :** D) Aluminium
Nickel, cobalt, and dysprosium are ferromagnetic materials (or exhibit similar strong magnetic ordering).
Aluminium is a paramagnetic material and is not ferromagnetic.

9. **Answer :** B) Non-magnetic material
Substances that are neither attracted by magnets nor can be converted into artificial magnets are classified as non-magnetic materials.

10. **Answer :** A) Mica
Mica is a mineral that does not respond to magnetic fields, making it a non-magnetic material.

11. **Answer :** B) Attract
A fundamental law of magnetism states that like poles repel each other, while unlike poles attract each other.

12. **Answer :** B) Soft iron
Temporary magnets are typically made of soft iron because it easily loses its magnetism once the external magnetic field is removed.

13. **Answer :** D) Steel
Permanent magnets are typically made from steel (or alloys like Alnico) because they retain their magnetism for a long time after being magnetized.

14. **Answer :** B) Temporary magnet
Temporary magnets, such as those made from soft iron, have very poor magnetic retention and lose their magnetism quickly.

15. **Answer :** B) Magnetic field
The space or region around a magnet where its magnetic influence or force can be detected is called the magnetic field.

16. **Answer :** A) Magnetic length $= \frac{5}{6} \times$ Geometric length
The standard relationship is that the magnetic length is approximately five-sixths of the geometric length.

17. **Answer :** A) Equatorial line
The imaginary line passing through the midpoint of the magnet and exactly perpendicular to its magnetic axis is called the equatorial line.

18. **Answer :** A) 15 cm
$L_m = \frac{5}{6} \cdot L_g$
$L_m = \frac{5}{6} \cdot 18$
$L_m = 15$ cm

19. **Answer :** C) Magnetic length
The straight-line distance between the North Pole and the South Pole inside a magnet is called its magnetic length.

20. **Answer :** C) Magnetic meridian
The imaginary vertical plane passing through the North and South Poles of a freely suspended magnet is called the magnetic meridian.

21. **Answer :** B) North pole
The end of a freely suspended bar magnet that naturally points towards the Earth's geographic North is called the North-seeking pole or simply the North pole.

22. **Answer :** B) Magnetic pole
The points situated slightly inside the ends of a bar magnet where its attractive power is maximally concentrated are called the magnetic poles.

23. **Answer :** C) Artificial Magnet
When a piece of iron is rubbed with a natural magnet like magnetite, it acquires magnetic properties and becomes an artificial magnet.

24. **Answer :** B) They can push or pull some objects without touching
A key characteristic of magnetic substances is their ability to exert non-contact forces (pushing or pulling) on other magnetic materials.

25. **Answer :** D) Magnetic sticker
Magnetic stickers (like fridge magnets) use permanent magnets, whereas devices like televisions, electric fans, and electric bells utilize electromagnets.

LEVEL-II

SINGLE CORRECT OPTION TYPE

1. **Answer :** C) Pins sticks at the ends of the magnet
The magnetic strength of a bar magnet is concentrated at its poles, which are located near the ends.
Therefore, the most pins will stick to the ends of the magnet.

2. **Answer :** B) Repulsion
Attraction can occur between a magnet and any magnetic material.
Repulsion can only occur between the like poles of two actual magnets, making it the only sure test of magnetism.

3. **Answer :** D) Iron
Iron is strongly attracted to magnets and is classified as ferromagnetic, not paramagnetic.

4. **Answer :** B) Closed curves
Magnetic lines of force are continuous, closed curves that travel from the North pole to the South pole externally, and from South to North internally.

5. **Answer :** B) Artificial
Artificial magnets, especially electromagnets and special alloys, can be made extremely powerful and molded into any desired shape.

6. **Answer :** A) Clockwise direction
The magnet will rotate to align its opposite poles with the adjacent magnets due to the forces of attraction and repulsion.

7. **Answer :** C) 6 cm

$$L_m = \frac{5}{6} \cdot L_g$$

$$5 = \frac{5}{6} \cdot L_g$$

$$L_g = 5 \cdot \frac{6}{5}$$

$$L_g = 6 \text{ cm}$$

8. **Answer :** D) B and C
The maximum attractive strength of any bar magnet is located exactly at its two poles (both the North Pole and the South Pole).

9. **Answer :** B) Huge and weak magnet
The Earth behaves like a massive magnet due to its molten core, but the actual strength of its magnetic field on the surface is relatively weak.

10. **Answer :** C) Weakest, strongest
The magnetic field lines are widely spaced at the center and densely packed at the poles.
Thus, the strength of a magnet is weakest in the middle and strongest at its ends.

11. **Answer :** B) 12cm

$L_m = \frac{5}{6} \cdot L_g$

$10 = \frac{5}{6} \cdot L_g$

$L_g = 10 \cdot \frac{6}{5}$

$L_g = 12$ cm

ONE OR MORE THAN ONE CORRECT OPTION TYPE

12. **Answer :** A, B, C

A magnet can lose its magnetic properties (demagnetisation) if it is heated strongly, hammered forcefully, or repeatedly dropped from a height.

13. **Answer :** A, C, D

Cobalt, iron, and steel are all strongly ferromagnetic materials, meaning they can be magnetized and used to make magnets.

Ebonite is non-magnetic.

14. **Answer :** B, C, D

A comb (plastic), a book (paper), and wood are all non-magnetic materials. They do not interact with magnetic fields and will not get attracted to a magnet.

15. **Answer :** A, B

Ferrite is widely used to manufacture powerful permanent magnets that are essential components in radios and transistors.

COMPREHENSION TYPE

16. **Answer :** B) North-South

A freely suspended magnet aligns itself with the Earth's magnetic field, coming to rest in the geographic North-South direction.

17. **Answer :** A) Geometrical North

The pole of the magnet that seeks or points towards the Earth's Geometrical North is defined as its North end.

18. **Answer :** B) Geometrical South

The pole of the magnet that naturally settles pointing towards the Earth's Geometrical South is defined as its South pole.

19. **Answer :** C) Diamagnetic materials

Materials that experience a weak repulsion from a magnetic field are classified as diamagnetic materials.

20. **Answer :** B) Iron nails

Iron is a ferromagnetic material, so the iron nails will be strongly attracted by the magnet.

Plastic, rubber (eraser), and wood are non-magnetic.

21. **Answer :** B) Magnetic material

Any material that experiences an attractive force when placed near a magnet is broadly termed a magnetic material.

MATRIX MATCHING TYPE

22. **Answer :** a → S, b → P, c → T, d → R, e → Q
A magnet curved so its poles face each other is a Horse-shoe magnet (S).
A closed continuous curve moving from north to south represents a Magnetic line of force (P).
The area around the magnet where its influence can be found is the Magnetic field (T).
A very small freely suspended needle used for tracing field lines is a Tracing compass (R).
A device used by sailors to find geographic direction is a Magnetic compass (Q).

23. **Answer :** a → Q, b → P, c → S, d → R
Neodymium makes the strongest artificial permanent magnets (Q).
Gold and bismuth are weakly repelled by magnets, making them Diamagnetic materials (P).
Nickel, cobalt, and dysprosium are strongly attracted, classifying them as Ferromagnetic materials (S).
Manganese, aluminium, and air are weakly attracted, classifying them as Paramagnetic materials (R).

ASSERTION AND REASON TYPE

24. **Answer :** A) Both A and R are correct and R explains A
Heating a magnet adds thermal kinetic energy which disturbs and randomizes the orderly alignment of its internal molecular magnets.
This directly causes the overall magnetism to reduce.

25. **Answer :** D) A is wrong but R is correct
The Assertion is false because the magnetic poles are not located *exactly* at the extreme ends, but rather slightly inside the magnet.
The Reason is factually correct as the attractive force is indeed maximum near the ends.

INTEGER TYPE

26. **Answer :** 4
The chemical formula for magnetite is Fe_3O_4.
Looking at the subscript for oxygen (O), it contains exactly 4 oxygen atoms.

27. **Answer :** 3
Based on their response to magnetic fields, materials are divided into 3 main types: Ferromagnetic, Paramagnetic, and Diamagnetic materials.

28. **Answer :** 2
Every magnet, regardless of its shape or how many times it is broken, always possesses exactly 2 poles (a North pole and a South pole).

PREVIOUS YEAR QUESTIONS

1. **Answer :** B) R
The magnetic field strength is always strongest directly at the poles of the magnet.
Position R is situated right at the pole, where the magnetic field lines are most densely concentrated.

2. **Answer :** → → (Arrows pointing away from North and towards South)
Magnetic field lines exit the North pole and enter the South pole.

3. **Answer :** C) I and III only
When a bar magnet (N-S) is broken into four pieces, each piece becomes an independent N-S magnet.
The poles will be: $A(N) - B(S)$, $C(N) - D(S)$, $E(N) - F(S)$, $G(N) - H(S)$.
Statement I is true: A, C, E, and G are all North poles (like poles).
Statement II is false: B, D, and H are South poles, but G is a North pole.
Statement III is true: C is a North pole and H is a South pole; unlike poles attract.

4. **Answer :** NS NS NS NS
When a single bar magnet is broken, new poles are immediately formed at the broken ends.
Each individual broken piece acts as a complete, independent dipole with its own North and South pole oriented in the same original direction.

